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Harnessing deep learning for air pollution forecasting: trends, techniques, and future prospects

Salvator Lawrence[1] · **Srimuruganandam Bhathmanabhan**[2]

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Abstract
Air pollution is a serious global public health threat arising from exposure to toxic ambient pollutants, including particulate matter (PM), sulphur oxides (SOx), nitrogen oxides (NOx), ozone (O_3), carbon monoxide (CO), and ammonia (NH_3). Traditional statistical and deterministic forecasting models often fail to adequately represent nonlinear interactions among multiple pollutants, meteorological drivers, and anthropogenic influences, motivating the growing adoption of deep learning (DL) approaches. This systematic review synthesizes evidence from more than 150 peer-reviewed studies conducted across diverse geographical regions and employing a wide range of DL architectures, including standalone, hybrid, and advanced spatiotemporal models. Using structured quantitative summaries, rank-based performance comparisons, and methodological assessments, the review identifies leading model families, analyzes pollutant- and horizon-specific performance trends, and evaluates robustness and generalizability across spatial and temporal contexts. Overall, DL models generally outperform traditional approaches, particularly when multisource inputs and spatiotemporal dependencies are explicitly modeled. Nevertheless, the literature remains fragmented, with a strong concentration of studies in data-rich urban regions of Asia, heterogeneous datasets, inconsistent evaluation protocols, limited transparency, and weak external validity. Addressing these limitations requires standardized preprocessing and benchmarking practices, improved explainability and uncertainty quantification, and the development of globally representative datasets. Emerging directions, including hybrid, physics-informed, and generative DL architectures, offer promising pathways to enhance reliability and operational deployment. Collectively, this review provides a comprehensive and critical synthesis of DL-based air quality forecasting, offering actionable insights for researchers, practitioners, and policymakers seeking transparent, generalizable, and policy-relevant prediction systems for environmental management and public health protection.

Keywords Air quality forecasting · Deep learning models · Spatiotemporal prediction · Hybrid architecture · Atmospheric pollution · Environmental monitoring

Extended author information available on the last page of the article

1 Introduction

Clean air is necessary for healthy living and is one of the most decisive factors of the overall quality of life (Manna and Anitha 2024). Ambient air is filled with a complex of toxic pollutants causing both acute and chronic health effects, and their role in causing global mortality is significant (WHO 2018). Recent estimates show that close to 99% of the world's population is exposed to polluted air with most of the burden falling on the low- and middle-income countries (WHO 2024). The negative health effects of criteria pollutants (i.e., particulate matter (PM_x), carbon monoxide (CO), sulphurous oxides (SO_x), nitrous oxides (NO_x), ozone (O_3), and volatile organic compounds (VOCs), as well as their respective source origins have been well-documented through epidemiological and toxicological studies (Diaz Resquin et al. 2023; Tuluri et al. 2023; Inam et al. 2024; Rahaman et al. 2025). Air pollution disproportionately impacts vulnerable communities, such as children, older people, people with pre-existing medical conditions and socioeconomically marginalised communities. The associated global economic burden is estimated to be around USD 8 trillion, which is more than 6.1% of the annual global gross domestic product (World Air Quality Report 2022). Collectively these environmental, health and economic consequences highlight the urgent need for sound air pollution mitigation strategies (Dey and Pal 2023; Inam et al. 2025e).

Accurate prediction of the ambient levels of pollution is a crucial factor in the prevention and control of air pollution (Zhang et al. 2022a; Sundaramoorthy 2024). Reliable air quality forecasting is dependent on the availability of quality data and powerful modelling frameworks. One of the main aims of air quality prediction is to give advance warning of prevailing and future pollution conditions to the public so that preventive measures can be taken in time during pollution episodes. Historically, air quality forecasts provided by traditional statistical and deterministic models have been the primary methods of air quality prediction. However, these approaches have frequently not been able to provide high-precision forecasts (Choudhury et al. 2022; Lin et al. 2022; Lee 2024), mainly because of their poor capacity to capture complex relationships among atmospheric pollutants, meteorological variables and anthropogenic activities (Chen et al. 2024a; Kalantari et al. 2024). Conventional models are also not able to handle multiple pollutants simultaneously leaving many of the atmospheric interaction mechanisms under-resolved (Rani Hemamalini et al. 2022). These limitations indicate the need for improved and more accurate forecasting methodologies.

The advent of artificial intelligence (AI) has been widely acknowledged as a transformational solution to a number of global problems including air quality management. AI-driven models have contributed substantially to improving the air quality prediction and have facilitated in making a sustainable and smart urban environment (Zhou et al. 2019). In particular, the combination of sensor networks and prediction models based on AI has been used increasingly for the monitoring and control of air quality in green cities (Zheng et al. 2023). A fundamental subset of AI, Machine Learning (ML), can make systems learn from historic data to forecast the air quality outcomes without the rules codes explicitly programmed (Lawrence and Bhathmanabhan 2025). Specialized branch of ML, Deep Learning (DL) has been specially developed for processing large scale datasets that are complex and unstructured and has improved levels of predictive accuracy. Mirroring the working of the human brain, DL models simulate interconnected structures in the neural network to model

data relationships that are complex. As a result, DL-based approaches are especially successful to model heterogeneous and nonlinear atmospheric processes (LeCun et al. 2015; Chatterjee et al. 2024; Vu et al. 2024; Wang et al. 2024b; Lin et al. 2025). As a result, DL has been identified to be one of the most reliable AI paradigms for air quality forecasting in the past years (Chang et al. 2020b; Wang et al. 2021; Li et al. 2025). These approaches have a critical role to play in the broader goal of advancing interdisciplinary approaches in research, supporting evidence-based policy making, and improving the accuracy of forecasts. Nevertheless, although they are becoming widely used, there are still large knowledge gaps in DL-based air pollution prediction research.

This review fills these gaps by systematically reviewing recent developments of standalone and hybrid DL models for air quality prediction. The study classifies various methodologies of DL, reviews the spatiotemporal modeling approaches for time series forecasting in global contexts, and finds the most commonly used and high-performing models using comparative analysis. The synthesized findings call for the potential of DL-based frameworks to support real-time environmental monitoring, public health protection, and climate change mitigation efforts. In order to further define the scope and novelty of the present review, we explicitly consider the fragmented nature of the existing research, for which studies differ considerably in terms of focus on the pollutants, forecasting time horizons, data sources, and DL architectures. Many previous reviews focus on a single DL model family, a particular pollutant or a small geographical area, thus providing incomplete or local information. In contrast, this review provides a global spatiotemporal modeling trends, robustness, practical deployment issues and cross-architecture performance comparisons around holistic synthesis. This global analytical approach makes the study an updated and comprehensive contribution in relation to those of the past.

Additionally, the review includes an orderly evaluation of model generalizability across regions, seasons, and monitoring stations, addressing directly issues concerning the external validity. The lack of external validity is a major limiting factor for DL-based air quality studies that have been published recently. Although there is no independent experimental validation, the review is a systematic synthesis and critical evaluation of existing literature based on structured quantitative analyses, rank-based performance comparisons, and methodological assessments. This approach provides a reproducible and analytically sound view on DL applications in air quality prediction and goes beyond descriptive summaries to produce actionable insights for researchers and practitioners.

2 Comparison with existing scoping reviews

To situate the present review within the broader literature and to clarify its relationship to earlier surveys, Table 1 offers a concise comparison with three closely related systematic or scoping reviews on AI and DL–based air quality forecasting. The comparison covers some key dimensions, including topical scope, pollutants considered, temporal coverage, modeling approaches, treatment of spatiotemporal dependencies, evaluation practices and approaches to synthesis and generalization. Rather than being a criticism of previous work, this comparison aims at making explicit how the current review fits in–and extends–existing efforts. Some amount of thematic overlap with previous reviews is both expected and unavoidable in such a rapidly growing field as DL-based air quality prediction. Previous

Table 1 Summary of existing scoping reviews and the incremental contributions of the present study

Aspect	(Liao et al. 2020)	(Liu et al. 2021)	(Chadalavada et al. 2025)	This Review
Publication year	2020	2021	2025	Under Review
Pollutants covered	Multiple pollutants	Multiple criteria pollutants	Multiple pollutants	Multiple criteria pollutants
Temporal coverage	Up to 2019	Up to 2020	Up to 2024	Up to 2025
Data sources	Single-city and regional datasets	Heterogeneous public datasets	Public monitoring networks	Multi-source, multi-region datasets
Model families	DL	Statistical, ML, and basic DL	ML and DL (broad)	DL, hybrid, graph-based models
Spatiotemporal modeling	Limited spatial modeling	Time-series dominant	Descriptive discussion	Formal spatiotemporal taxonomy
Evaluation depth	Metric reporting	Metric-level comparison	Qualitative comparison	Quantitative rank-based synthesis
External validity analysis	Largely absent	Limited	Narrative only	Systematic generalization analysis
Quantitative synthesis	None	None	None	Median- and rank-based aggregation
Practical/deployment focus	Minimal	Minimal	High-level	Deployment and policy-oriented

This table compares the present review with three closely related systematic and scoping reviews, highlighting key dimensions such as scope, pollutants, modeling approaches, spatiotemporal treatment, evaluation depth, and synthesis strategy.

The final column emphasizes the unique contributions of this study, including rank-based quantitative analysis, a formal spatiotemporal taxonomy, and systematic evaluation of external validity across geographical, seasonal, and station-level settings

reviews have played important roles in collating emerging evidence, clarifying the common modelling paradigms, and summarizing performance trends across datasets and pollutants. The current study provides a direct extension of this work, drawing on the previous studies, but with a different focus on being more integrative and analytical.

In comparison to former reviews, the current work gives more emphasis on structured comparison and transferability. It gives an introduction to a rank-based quantitative synthesis among model families, proposes a formal taxonomy of spatiotemporal architectures, and a systematic examination of evidence for external validity across geographical regions, seasons, and monitoring stations. In addition, the review explicitly connects modeling choices to deployment considerations and policy-relevant use cases, turning away from descriptive summaries to a synthesis conducive to real-world interpretation as well as decision-making.

3 Methodology framework

The systematic review was conducted in accordance with the guidelines for evidence synthesis practices and was based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework (Haddaway et al. 2022). The methodological protocol has been developed in order to be transparent, reproducible and holistic in covering existing research in DL methodologies into air quality prediction and analysis. Although the protocol was not registered in any registry all the methodological procedures such as search, strategy, screening workflow and extraction processes were fully documented to the maintenance of methodological consistency.

3.1 Review protocol summary

The systematic review followed a predefined, PRISMA–aligned methodological protocol to ensure transparency and reproducibility. For clarity and ease of replication, the key design choices of the review are summarized below. Detailed descriptions of each component are provided in the corresponding subsections (Sect. 3.2–3.7).

- *Databases searched*: ScienceDirect, Scopus, and Google Scholar.
- *Search period*: 2019–2025 (last updated on 1 December 2025).
- *Search strategy*: Boolean combinations of air-quality and DL–related terms (see Sect. 3.3 for full search strings).
- *Study selection*: Peer-reviewed English-language journal articles.
- *Screening process*: PRISMA-guided identification and screening with title–abstract and full-text review stages; dual-reviewer screening was adopted to reduce subjective bias (Sect. 3.4).
- *Inclusion criteria*: DL–based air-quality or AQI forecasting studies, including standalone and hybrid models with spatiotemporal considerations (Sect. 3.5).
- *Exclusion criteria*: Studies limited to non-DL machine learning methods, non-forecasting tasks, or insufficient methodological transparency (Sect. 3.5).
- *Data extraction*: Standardized extraction of datasets, pollutants, model architectures, validation strategies, evaluation metrics, and deployment considerations (Sect. 3.6).
- *Risk-of-bias assessment*: Model-specific quality evaluation adapted from TRIPOD-AI

and PROBAST domains, with studies classified into low, moderate, or high-risk categories (Sect. 3.7).
- *Handling of inaccessible full texts*: Articles that could not be accessed due to subscription restrictions or unavailable full texts were excluded after screening (Sect. 3.4).

3.2 Search strategy

A structured and systematic search was conducted in three major scientific databases including ScienceDirect, Scopus and Google Scholar. These databases were chosen because of their comprehensive indexing of research in the areas of artificial intelligence, environmental sciences and atmospheric modeling. The search timings were set from (2019–2025) to reflect the high speed in which DL approaches have developed and the subsequent widespread adoption in air quality forecasting.

3.3 Search strings

Search strings incorporated both environmental and DL-specific terms, using Boolean operators to maximize retrieval sensitivity. The main search string applied was: ("Air quality predictions" OR "Air Pollution Forecasting" OR "Air Quality Index Forecasting") AND ("Deep Learning" OR "Neural Network" OR "Convolutional Neural Networks" OR "CNN" OR "Long Short-Term Memory" OR "LSTM" OR "Recurrent Neural Networks" OR "RNN" OR "Gated Recurrent Unit" OR "GRU" OR "Deep Belief Network" OR "DBN" OR "Sparse Autoencoder" OR "SAE" OR "Deep Neural Network" OR "DNN" OR "Multi-Layer Perceptron" OR "MLP" OR "Generative Adversarial Network" OR "GAN" OR "SpatioTemporal Deep Learning"). The search was last updated on 1 st December 2025. Only peer-reviewed, English-language journal articles were included to ensure scientific reliability.

3.4 Screening workflow and PRISMA counts

A PRISMA-style workflow (Fig. 1) was followed in order to systematically identify relevant studies and screen them for relevance. The initial search gave a yield of 517 records from electronic databases and instructional library sources. After excluding 83 duplicate records, 434 articles were taken to title-abstract screening.

Based on predefined relevance criteria, 98 articles were excluded because they only addressed ML techniques or they were unrelated to atmospheric studies. The remaining 336 articles were assessed for full-text retrieval. However, 106 articles could not be accessed because of subscription restrictions, paywalled content and unavailability of full texts. The remaining 230 articles went through a full-text assessment, during which 71 articles were excluded for different reasons, such as lack of focus on forecasting, lack of methodological details, or being non-archival conference publications. Ultimately, 159 studies met all of the inclusion criteria and were included in the final synthesis. To improve the reproducibility and minimize the subjective bias, a dual review screening strategy was adopted. Two reviewers independently screened articles, and, disagreements were resolved by consensus.

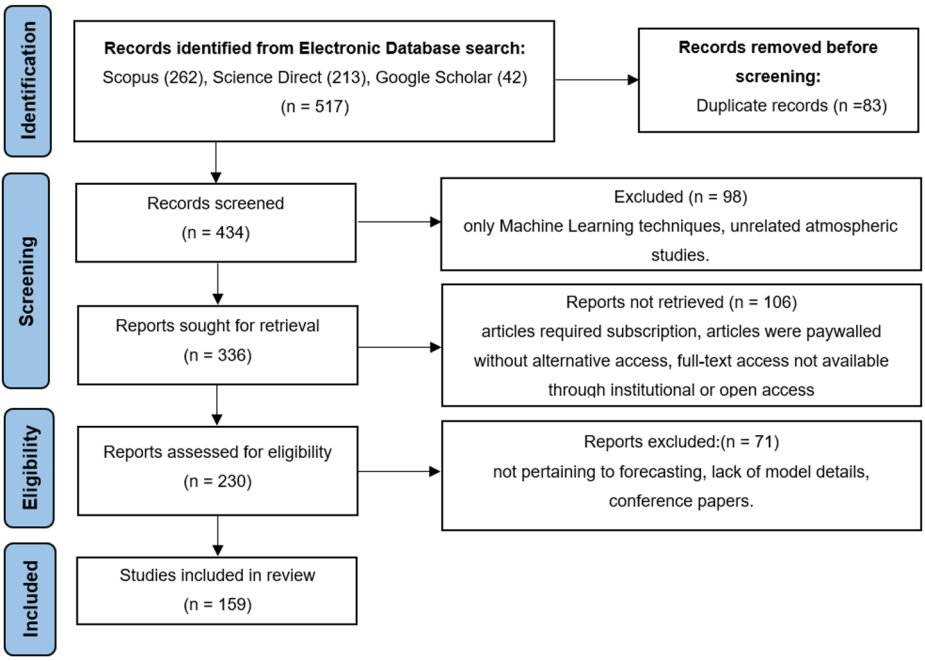


Fig. 1 PRISMA 2020 flow diagram illustrating the complete process of study identification, screening, eligibility evaluation, and final inclusion for this systematic review. The diagram ensures transparency and reproducibility in the selection of literature related to Deep Learning approaches for air quality forecasting

3.5 Inclusion and exclusion criteria

Studies were included if they met the following criteria: Air-quality prediction or forecasting using at least one DL model, Spatiotemporal ambient air-quality forecasting, Standalone or hybrid DL model for Air Quality Index (AQI) or pollutant concentration prediction, Forecasting of single or multiple pollutants and Peer-reviewed journal articles. Exclusion criteria were: Studies that only used ML approaches with no DL component; DL studies that are beyond the scope of air-quality applications; Articles that were not characterized by an adequate level of methodology transparency or empirical validation; Studies that did not relate to AQI or pollutant-level forecasting.

3.6 Data extraction and synthesis

A standardized schema of data extraction was developed to ensure comparability among studies. Some extracted fields were: Study location and year, Dataset type, source, and variables, Pollutants forecasted, DL model architecture, Input features, Training/Validation approach, Evaluation metrics, and Key findings, limitations, and deployment considerations. This structured extraction enabled a consistent, thematic, architectural and performance-based comparison between the included studies. As a part of the data extraction process, all

studies were also categorized depending on their spatiotemporal modeling strategies, this is discussed in detail under Sect. 6.

3.7 Study quality and risk of bias (RoB) assessment

To ensure reliability of the synthesized evidence a modelling specific quality assessment was conducted for all included studies. Due to the lack of a universal risk of bias tool for DL-based environmental modelling, a simplified checklist was created based on key aspects from TRIPOD-AI and PROBAST guidelines as an adaptation of the original tools. Studies were reviewed on seven key areas; completeness of dataset, standardization of data & bench-marking, transparency of model architecture, reporting of hyperparameters, validation strategy, baseline & reproducibility, and data leakage & overfitting issues. Each of the domains was rated as Low, Moderate or High risk based on the completeness of information reported. To ensure consistency and transparency, the ratings of the domains were operationalized on a three-level scoring scale (Low = 0, Moderate = 1, and High = 2) and an Overall Risk Average was calculated by taking the average of all domain scores. Studies were classified according to predefined values: Low risk (0–0.67); Moderate risk (0.68 to 1.33); High risk (1.34–2.00). A summary of the assessment domains is presented in Table 2, whereas the detailed ratings of all 159 studies are presented in Supplementary Table ST1.

4 Air quality predictions

Air quality prediction is an estimate of probable current values of pollutant concentrations from available sources of information such as emission sources, meteorological information, and traffic-related factors. In contrast, air quality forecasting can be described as the estimation of ambient air conditions at some future time with reference to historical observations. Accurate prediction and forecasting of air pollution are likely to be necessary for giving advance warnings to the public before episodes of pollution occur. Such timely alerts form part of better public health outcomes, economic growth, sustainable development and environmental research.

4.1 Conventional methods to predict air quality

Conventional air quality prediction methods are based mostly on statistical models and deterministic models which are often expensive, restricted in availability and cannot work well in practice (Lin et al. 2021). These models struggle when used for complex datasets, and also have inconsistent performance because air pollution processes are of an inherently dynamic and nonlinear in nature (Masood and Ahmad 2021). Common statistical techniques, such as analysis of variance (ANOVA), correlation analysis and regression models, are usually applied to determine the relationships between meteorological variables and emission data. Deterministic models such as Eulerian and Lagrangian models are based on the representation of the physical and chemical processes. However, these approaches are facing serious challenges from fragmented monitoring networks, simplified numerical formulations, as well as uncertainty used from chemistry-transport modeling (Liao et al.

Table 2 Summary of Risk-of-Bias (RoB) assessment domains for included studies

ROB Domain	Description	Assessment Criteria	Low-Risk indicators	High-Risk indicators	Risk Categories
Data Completeness	Transparency and adequacy of dataset description	Clear dataset metadata, missing data handling	Full description of data sources, period, pollutants, preprocessing	Missing dataset description, unclear handling of missing values	Low/Moderate/ High
Data Standardization & Benchmarking	Adequacy of data preparation practices for cross-study comparability	Benchmark usage or datasets justification; preprocessing transparency; data imbalance consideration.	Benchmark or justified datasets with documented preprocessing; imbalance addressed	No benchmark rationale; unclear preprocessing; imbalance ignored	Low/Moderate/ High
Model Architecture Transparency	Clarity of DL model description	Architecture, layers, fusion/ spatiotemporal design	Fully specified architecture with diagrams	High-level or incomplete architecture descriptions	Low/Moderate/ High
Hyperparameter Reporting	Reporting of training configuration	Learning Rate, batch size, epochs, optimizer	Full hyperparameter table, tuning strategy	No hyperparameters reported	Low/Moderate/ High
Validation Strategy	Methodological correctness of evaluation	CV, temporal splits, leakage checks	Proper temporal CV, multi-site tests	Random splits, improper leakage handling	Low/Moderate/ High
Baseline & Reproducibility	Strength of comparative design	ML baselines, simpler DL baselines	Multiple valid baselines, reproducibility artifacts	No baselines or unverified claims	Low/Moderate/ High
Data Leakage & Overfitting issues	Leakage prevention &model generality	Scaling order, splits, regularization	Preprocess after splitting, no shuffle, regular/early stop	Scaling before split, shuffle leakage, no overfit checks	Low/Moderate/ High

2020). Consequently, the effort towards creating accurate, robust and intelligent modeling frameworks is now essential for the correct prediction of air pollution (Menares et al. 2021).

4.2 Advanced methods to predict air quality

AI has been applied in different areas, but it has been particularly useful when it comes to the field of environmental management. AI-based modeling strategies make it possible to better predict air pollutant concentrations (Liu et al. 2021). Subfields of AI, specifically ML and DL, have enabled the means to develop sound models to predict air quality. Among these DL approaches have shown better performance as compared with the traditional ML approaches in forecasting air pollution levels (Pande et al. 2024). Conventional modeling methods can often not deal well with large quantities of data and data that are highly complex and cannot capture spatial dependencies within atmospheric systems (Yang et al. 2024a). In comparison, DL models are basically good at the study of complex interactions between the atmosphere and the processing of high-dimensional and non-linear data, which led to more accurate forecasts of air quality (Zhang et al. 2020a; Choudhury et al. 2022; Chen 2024; Chen et al. 2025a). The consistency and reliability of DL-based modeling approaches also provide valuable information in terms of guiding emissions control policies which are aimed to protect public health.

5 Concept of deep learning

DL is a method of utilizing neural networks with many layers to learn inherent patterns in data and make correct predictions (Hanigan et al. 2025). It has also proven to be very effective in applications of time series prediction (Omri 2024; Jiménez-Navarro et al. 2024). DL models have immense potential for dealing with complex and high dimensional datasets (Moursi et al. 2021; Abbas and Raina 2023; Duan et al. 2023). Unlike the divide-and-conquer paradigm traditionally followed in ML approaches (Chang et al. 2020a), DL is an end-to-end learning approach (Abirami and Chitra 2021), where more than one weighted layer is optimized simultaneously, instead of one after the other. DL architectures are comprised of a sequence of interconnected layers, nodes, trainable weights, hence they are very complicated structures, comparatively, to conventional ML models. In addition, DL models can be trained to learn directly from unstructured data, whereas ML techniques typically only take structured inputs (Liao et al. 2020). A major feature of DL is that they can capture nonlinear relationships between input and output variables (Njaime et al. 2024).

DL is now widely applied in air pollution engineering where models are used for prediction, forecasting and classification tasks (Moursi et al. 2021). Although the variability of the pollutants and the management of large datasets are challenges for accurate forecasting, DL approaches handle and solve the issues effectively by efficiently managing large datasets and learning complex patterns to give reliable results (Zeng et al. 2022). Ongoing improvements in DL modeling are making it possible to continuously and iteratively simulate in a way that cannot be done with human ability to manually analyze and interpret such a large and complex data set. Apart from regression and classification algorithms, DL algorithms can also be used to detect pollution hotspots and potential sources of emission (Abirami and Chitra 2021). A variety of DL architectures have been used for ambient air quality prediction

and forecasting. Table 3 gives an overview of the major DL and hybrid models involved in air quality forecasting, by providing a summary of their functional roles, architectural components, strengths and limitations.

5.1 DL techniques in AQ predictions

Air quality prediction studies make use of a broad spectrum of DL model families, including recurrent architectures (RNN, LSTM, GRU), convolutional neural networks (CNNs), feedforward models (MLP, DNN, DBN), and hybrid or generative approaches such as CNN–LSTM, SAE, and GAN. Within each model family, there are unique methodological invariant strengths. CNNs are especially effective for collecting spatial relationships among pollutants distributions, whereas recurrent architectures are widely applied for modelling temporal dependencies, and feedforward networks are often applied for representing nonlinear relationships among input variables in the time series forecasting task (Liao et al. 2020). Hybrid models combine complementary aspects of spatial and temporal to increase the predictive capability (Che 2024). Comparative evidences have been found with LSTM based architectures and a hybrid architecture of CNN and LSTM consistently exhibit high performance in short term air quality forecasting, while simpler MLP and DNN architectures tends to exhibit high variability on different locations and on forecast horizon with respect to being. Moreover, the inconsistent reporting of data preprocessing procedures and normalization strategies persists and continues to challenge the ability to compare cross-study; highlighting the need for standardised and transparent evaluation protocols in this field. The following representative studies are discussed to highlight the comparative performance and applications and are not meant to provide a comprehensive listing of all models available:

5.1.1 Convolutional networks (CNNs)

CNNs are particularly effective in learning spatial pollutant structures. They are widely used when pollutant information is arranged in grid-like structures including, satellite images, sky photographs, or spatial sensor networks (Dey 2024). Studies have shown strong predictive accuracy by leveraging CNNs' ability to capture regional pollutant interactions. Chinese research acquired data from Aotizhongxin station that showcased a strong association of $PM_{2.5}$ with R^2 value of 0.80 among its neighbouring stations (Bekkar et al. 2021).

5.1.2 Recurrent networks (RNN, LSTM, GRU)

The LSTMs address the limitations of traditional RNNs by incorporating memory cells and gating mechanisms, allowing them to retain long-term temporal information. They remain among the most effective models for pollutant time-series prediction and have consistently achieved high accuracy: for example, A Turkish study reported that the LSTM algorithm outperformed all other models for forecasting $PM_{2.5}$ and SO_2 levels (Das et al. 2022).

GRUs simplify LSTM's structure while retaining the ability to model long-term dependencies, making them computationally effective alternatives. They have shown strong performance even with limited datasets. A federated Learning based Bidirectional Gated Recurrent unit (FL-based BGRU) was applied to predict $PM_{2.5}$ and found optimum results comparing the traditional ML models (Dey and Pal 2022).

Table 3 Overview of key deep learning and hybrid models for air quality forecasting: functions and limitations

Model	Functions	Limitations	Reference
STHTNN	• Combines short-term spatiotemporal features with long-term periodic patterns for improved air quality prediction.	• Longer training and inference times compared to baseline models, which may limit scalability and real-time application.	Chen et al. 2024a, b, c, d
LSTM & GRU	• Mitigate vanishing gradient issues, enabling effective modeling of complex temporal dynamics.	• Scaling to large datasets is challenging, requiring careful management of computational resources and optimization strategies.	Chiang et al., 2024
LSTM-BNN	• Integrates single-step recursive LSTM forecasting with Bayesian reasoning for data fusion, combining observational and numerical estimates.	• Integrating predictions from CCAM-CTM into the recursive forecast remains challenging.	Nguyen et al., 2023
ARIMA-CNN-LSTM	• Decomposes data into linear and nonlinear components; the nonlinear part is processed via DNN to generate predictions, combining both aspects for final output.	• Incorporating multiple influencing factors is complex, requiring careful handling of nonlinear interactions.	Duan et al. 2023
CNN –LSTM/NARX	• Uses NARX to select historical $PM_{2.5}$ inputs for CNN-LSTM; CNN extracts 4D features, LSTM captures temporal dynamics, followed by tanh and linear layers for final output. The output from the LSTM network is then reduced through tanh dense layer, followed by a linear dense layer to generate the final $PM_{2.5}$ output.	• NARX integration adds complexity, potentially affecting stability, convergence, and accuracy.	Moursi et al., 2021
LSTM & DFFNN	• Forward propagation computes loss; backpropagation obtains gradients; LSTM captures synoptic patterns and predicts pollutants based on meteorological inputs.	• Integrating spatial features is challenging, especially for capturing heterogeneity and scaling to large spatial datasets.	Menares et al. 2021
CNN-LSTM	• CNN extracts salient features, flattened into 1D tensor for LSTM; reshaped to original structure before output generation.	• Excludes transboundary pollutant contributions, which may significantly affect local air quality.	Bekkar et al. 2021
CNN	• Feature extraction: captures local characteristics and positional relationships. Feature mapping: uses weight sharing to improve efficiency and reduce parameters.	Limited in predicting peak values; struggles with complex terrains; currently predicts single-point gas emissions; requires diverse training samples for broader applicability.	Ni et al., 2020

STHTNN Spatiotemporal hierarchical transmit neural network, *GRU* Gated recurrent unit, *LSTM-BNN* Long short-term memory Bayesian neural network, *ARIMA-CNN-LSTM* Autoregressive integrated moving average – Convolutional neural network - Long short-term memory, *NARX* Nonlinear autoregression with exogenous input, *DFFNN* Deep feedforward neural network

5.1.3 Feedforward networks (MLP, DNN, DBN)

These models are effective in handling nonlinear feature interactions. MLPs, as simple feedforward networks remain relevant due to their ability to approximate non-linear pollutant interactions. They are often used in hybrid workflows or as benchmark models. DBNs use stacked Restricted Boltzmann Machines for hierarchical representation learning. They are particularly useful for unsupervised feature extraction before supervised refinement. Studies in China highlight their strong performance, showing improved accuracy deeper DBN structures are used (Li et al. 2019; Tian et al. 2022). DNNs consist of multiple fully connected layers that capture non-linear relationships in pollutant data. They are effective general-purpose models and frequently serve as baselines for hybrid architectures. Their usefulness is demonstrated in tasks ranging from comprehensive air quality index (CAQI) prediction in South Korea (Kim et al. 2022) to multiregional pollutant mapping in China (Zou et al. 2021). Performance often rivals advanced sequence models when data is rich and well-structured.

5.1.4 Generative models (SAE and GAN)

SAEs compress high-dimensional data into meaningful latent representations before reconstruction and are thus useful for feature extraction and anomaly detection. A research in China put into operation the LSTM-SAE approach in air quality prediction with a classification accuracy of 91.22%, and with an error rate of 0.46 (Liu and Zhang 2021). GANs are used to synthesize data in a generator-discriminator network and are especially useful for data augmentation and imputing missing observations. A case study in North Macedonia applied a GAN model for air quality forecast at a regional level, having an accuracy of 0.88 (Kalajdjieski et al. 2020).

5.2 DL-network architecture

DL network architectures typically follow the form of an artificial neural network (ANN) which is a series of input, hidden and output layers. Meteorological and pollutant variables are fed in at the input layer and are then passed through one or more hidden layers which consist of interconnected neurons. Within these layers, learnable weights and activation functions are used in order to transform the inputs into higher-level representations of features (Rad et al. 2025). The output layer then produces the air quality values to be predicted. Figure 2 shows this end-to-end workflow, which includes both, basic neural network-based DL architectures and sophisticated hybrid modeling frameworks. Collectively, these components make the development of more robust and accurate spatiotemporal air quality forecasting systems possible.

5.3 DL for air quality forecasts

The inclusion of advanced neural network architectures via DL has resulted in a paradigm shift in air quality prediction, with more reliable and accurate predictions of air pollutant concentrations. Key applications are the analysis of spatiotemporal relationships and the prediction and forecasting of air pollutants. Despite continued challenges (including

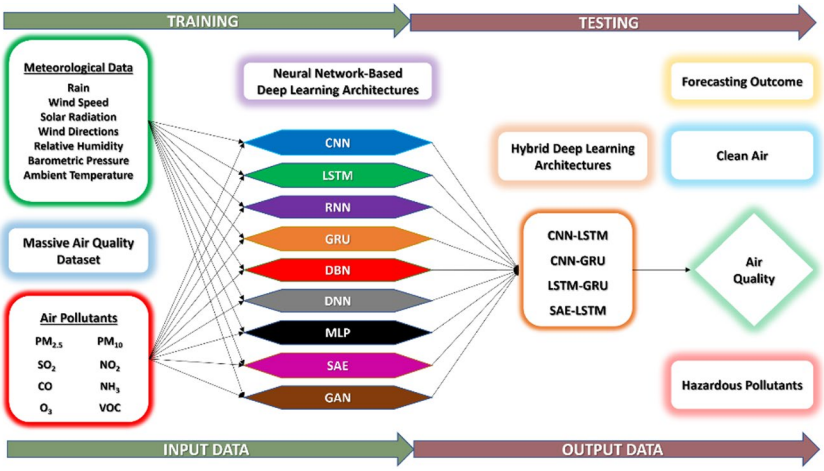


Fig. 2 Neural Network-based and Hybrid Deep Learning Model Architecture for Air Quality Forecasting. Key components such as input features, hidden layers, fusion pathways, and forecast outputs are depicted to highlight model functionality and comparative design principles

data quality and availability, model interpretability, intrinsic nonlinearity, integrating multi-source and multi-scale data, etc.), DL approaches have been shown to have great capability in addressing these challenges. Continued advances in DL have led to an increase in prediction accuracy, an ability to effectively process large amounts of unstructured data with complex dependencies, and to an increase in robustness to missing data, which has enabled the development of applications for real-time forecasting. In environmental studies, the measurement of a pollutant's size, and consequently particles which are 2.5 microns in diameter or less in the pollution Chamber, make $PM_{2.5}$ a fundamental marker for Air Quality evaluation. Numerous studies highlight the significance of concentrations of $PM_{2.5}$ because of their large effects on public health, economy and social development. However, the pronounced temporal variability of $PM_{2.5}$ remains difficult to model because of its strong nonlinearity and the resulting instability in air quality dynamics (Zhang et al. 2021).

In reviewing the literature, LSTM is the most widely adopted temporal encoder for modeling sequential pollutant patterns, frequently combined with CNN layers to capture local spatial structures (Bekkar et al. 2021; Eum and Yoo 2022). An emerging subset of research employs graph-based architectures to explicitly model monitoring networks as spatial graphs (Muthukumar et al. 2022b; Chen 2024).

Most graph-based models construct adjacency matrices based on geographic proximity or physical station connections, though only a few explore dynamic, learner or attention-guided graph structures (Yang and Zhang 2023; Li and Jiang 2023). Advanced architectures often integrate fusion mechanisms including attention layers (Huang et al. 2021; Zhou et al. 2022), multi-branch CNN-LSTM (Zhu and Xie 2023; Dey 2024), and encoder-decoder frameworks (Zhang and Gan 2023; Asaei-Moamam et al. 2023) to combine spatial and temporal information within a single model for improved spatiotemporal forecasting. Despite these innovations, comprehensive ablation studies or detailed component-wise evaluations remain rare, leaving gaps in methodological transparency. Overall, while spatiotemporal DL models demonstrate substantial diversity and sophistication, reporting standards and depth

of methodological detail vary considerably across studies, underscoring the need for more consistent and transparent evaluation practices in future research.

5.4 Global air quality studies

The geographical distribution of published articles about air quality prediction in different countries with DL approaches is shown in Fig. 3. Further, the performance of state of the art models in DL for individual air pollutants and air quality is summarised in this section. Table 4 gives a brief summary of the models used, the target pollutants, the geographical regions considered, the evaluation measures used and the main results reported in each work. The table covers many DL architectures from sequence-based model such as Long Short-Term Memory (LSTM) & Gated Recurrent Unit (GRU) to hybrid frameworks, CNN-LSTM (Li et al. 2020), advanced architectures such as spatiotemporal architectures such as GCN-ConvLSTM (Muthukumar et al. 2022a), DCRNN (Wang et al. 2022a), and GRU-GCN (Chen et al. 2024c).). It additionally contains more recent ways of doing this including Temporal Convolutional Networks (TCNs) (Wang et al. 2024a), and modern temporal convolution architectures (Li et al. 2025), in addition to attention-based models. Collectively, these studies have demonstrated the extensive variety of ways in which different family of DL models have been adapted to make $PM_{2.5}$, PM_{10}, gaseous pollutants and Air Quality Index (AQI) predictions for a wide range of environmental conditions. By reporting model architectures as well as the model outcomes helps to compare model effectiveness under various conditions, the evolution of beliefs in modeling strategies by myself and the repeatable strengths and shortcomings.

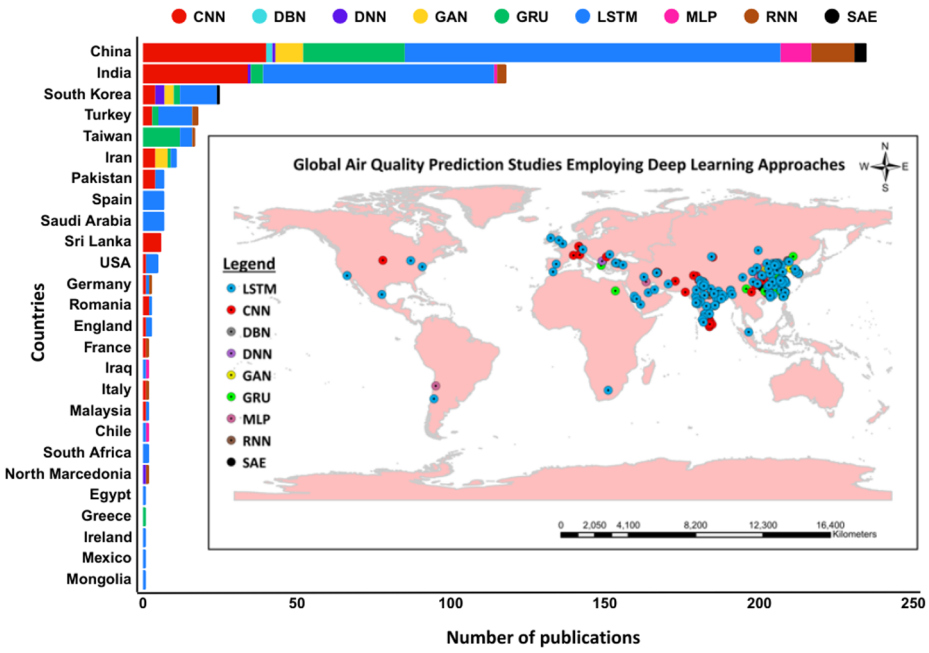


Fig. 3 Global geographical distribution of studies employing Deep Learning techniques for air quality prediction

Table 4 Overview of major deep learning Approaches, target Pollutants, evaluation Metrics, and key outcomes in air quality prediction

Deep Learning Approaches	Target Prediction	Study Area	Year	Evaluation Criteria	Outcomes	Author
DM-LSTM	$PM_{2.5}$, PM_{10}, and NO_x	Taipei City, Taiwan.	2019	RMSE, and MSE	• Improves spatio-temporal stability and multi-step forecasting accuracy for $PM_{2.5}$, PM_{10}, and NOx.	Zhou et al. 2019
TS-LSTME	$PM_{2.5}$	Jing-Jin-Ji, China	2021	RMSE, MAE, and R^2	• Significantly enhances $PM_{2.5}$ prediction performance.	Mao et al. 2021
CNN-LSTM	$PM_{2.5}$	Taiyun, China	2020	RMSE, MAE, and R^2	• Achieves high accuracy in $PM_{2.5}$ concentration prediction.	Li et al. 2020
ALSTM	$PM_{2.5}$	Taiwan	2020	MAE, RMSE, and MAPE.	• Effectively improves $PM_{2.5}$ prediction accuracy.	Chang et al. (2020)
ODMSCNN-LSTM	$PM_{2.5}$, PM_{10}, NO_2, CO, O_3, and SO_2	Xi'an, China	2021	MAE, RMSE, and MAPE	• Provides strong predictive performance for multiple pollutants.	Dai et al. 2021
MLEGRU	$PM_{2.5}$	Taiwan	2021	MAE, and RMSE.	• Outperforms other ensemble learning methods for $PM_{2.5}$ forecasting.	Lin et al. 2021
P-LSTM	$PM_{2.5}$	Delhi, Bengaluru, Chennai, Haryana, Jodhpur, Lucknow, India	2021	RMSE, MAE, MAPE and R^2	• Outperforms existing models across diverse evaluation metrics.	Aggarwal and Toshniwal 2021
MI-LSTM-MVO	SO_2, and NO_2	Kerman, Iran	2021	RMSE, MAE, and MAPE	• Shows higher accuracy than benchmark models considering varied input variables.	Heydari et al., 2022
DCRNN	O_3, and $PM_{2.5}$	Yangtze River Delta, China	2022	Accuracy, MAE, RMSE, and R	• Improves forecasts of O_3 and $PM_{2.5}$.	D. Wang et al. (2022)
GCN-ConvLSTM	$PM_{2.5}$	Los Angeles, USA	2022	RMSE, and NRMSE	• Enhances $PM_{2.5}$ prediction over existing methods.	Muthukumar, Cocom, et al. (2022)
Bi-LSTM	$PM_{2.5}$	Taiwan	2022	MAPE, RMSE, MSE, and MAE	• More suitable than other neural networks for $PM_{2.5}$ prediction.	Kristiani et al., 2022
WPT-LSTM	$PM_{2.5}$	Qingdao, China	2023	RMSE, NSE, and R^2	• Achieves state-of-the-art $PM_{2.5}$ prediction performance.	Zheng et al. 2023
MIA-LSTM	$PM_{2.5}$	Beijing, China & Ghaziabad, India	2023	RMSE MAE and R^2	• Improves accuracy of $PM_{2.5}$ forecasting.	Narkhede et al., 2023
GATBL-Learning	AQI	China	2023	MSE, RMSE and MAE	• Improves air quality index forecasting accuracy.	Zhang et al. (2023)

Table 4 (continued)

Deep Learning Approaches	Target Prediction	Study Area	Year	Evaluation Criteria	Outcomes	Author
TLS-BDLSTM	$PM_{2.5}$, and PM_{10}	Kolkata, India	2023	MAE, and RMSE	• Outperforms other methods in single- and multi-step forecasts.	Dutta and Pal, 2023
TCN	SO_2	Yunnan, China	2024	RMSE, MAPE, and R^2	• Produces improved SO_2predictions.	Q. Wang et al. (2024)
GRU-GCN	CO_2	(Yangtze River Delta, Pearl River Delta & Beijing–Tianjin–Hebei region), China	2024	MAE, RMSE, and MAPE	• Superior forecasting for single- & multi-step spatiotemporal CO2 estimates.	Chen et al. 2024a, b, c, d
Modern TCN	O_3	Los Angels, USA & Wuxi, China	2025	RMSE, MAE, sMAPE, and R^2	• Effectively extracts spatial dependencies and key aspects of ozone formation.	Li et al. 2025

To give an exhaustive overview of the reviewed literature, Supplementary Table ST2 summarizes all of the studies included in this work in a systematic way. Each publication is classified based on DL approach, target pollutant/prediction task, geographical study area and year of publication. The table further presents the corresponding prediction performance and evaluation metrics, which allows to consistently compare the results of different methodological choices and forecasting results. Together, this structured synthesis offers a clear picture of the current research trends with a regional application pattern and the big picture of the performance in DL models in air quality predictions.

Supplementary Table ST3 summarizes in detail the performance improvements reported in the literature by consolidating each DL approach with the information related to the prediction target, the documented degree of improvement, the evaluation criteria and reference sources. This structured collection is appropriate for the direct comparison of model performances for different forecasting tasks and different assessment protocols. By having the ability to depict such methodological and performance characteristics in a unified way, the table would allow for a transparent interpretation of the predictive gains and evaluation practices revealed in current DL-based air quality studies.

Supplementary Table ST4 further enriches this synthesis by putting together a detailed inventory of the recent DL models that have been applied to air quality prediction. Each of the studies will be classified with respect to model type, target pollutant or forecasting task, geographical study domain, year of publication, and reported predictive performance. The table also standardizes evaluation metrics employed across studies for consistent comparison of methodologic choices/predictive efficacy. This consolidated summary not only highlights the diversity of modelling strategies, but also the regional contexts where they have been deployed to offer a basis in identifying research trends, gaps and opportunities in the future.

Figure 4 shows the worldwide distribution of DL models used for air quality prediction and forecasting for the reviewed studies. $PM_{2.5}$ is the most commonly forecasted pollutant, and is considered in almost every type of model, thus highlighting the importance of the pollutant in air quality assessment. The bar chart shows the number of model types, such as recurrent, feedforward and hybrid models, and therefore highlights model popularity and adoption trends, as well as the relative focus on temporal and spatial forecasting approaches. The results of the analysis showed that the most common models used to predict air quality are LSTM-based models, which are suggested to be effective and powerful in predicting time-dependent data, i.e. forecast models. $PM_{2.5}$ becomes the most popularly sought pollutant, probably because of its severe effects on human health and the environment, though the prediction of AQI is also widespread, because it is an aggregate measure of the overall air quality. Amongst hybrid DL approaches, CNN-LSTM models are the most popular and their performance is quite good across studies. Overall, the reviewed air quality prediction efforts focus primarily on the seven criteria pollutants and especially on $PM_{2.5}$ and PM_{10}.

Figure 5 shows the doughnut charts for the percentage distribution of DL models used for forecasting individual criteria pollutants. These visualizations show how often each of the pollutants is considered for DL-based forecasting studies, giving an insight into current research priorities, data availability and possible gaps in environmental monitoring coverage. Overall, the results show that the LSTMs models followed by CNN models are the most applied models for the prediction and forecasting of major air pollutants.

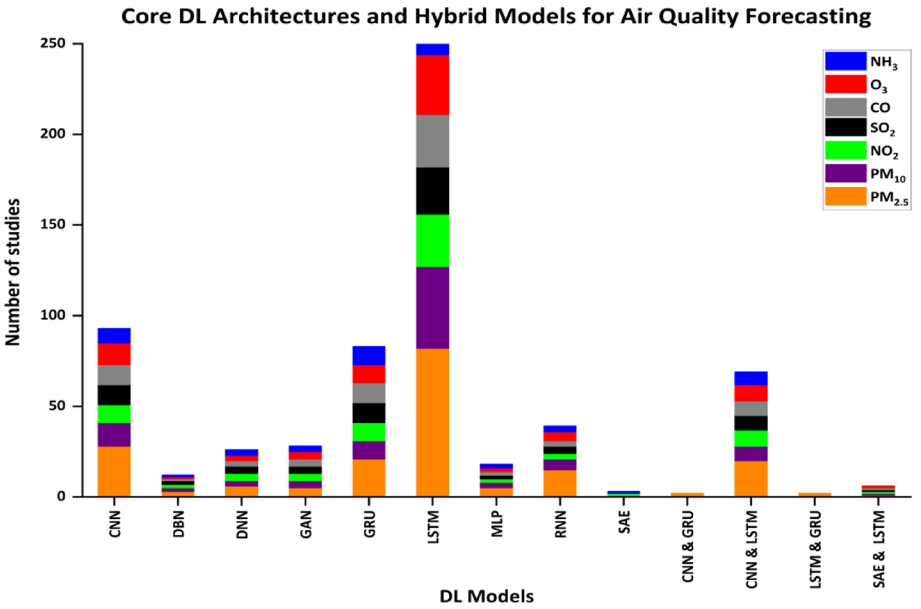


Fig. 4 Number of studies employing distinct core Deep Learning architectures and their hybrid variants for air quality prediction across the reviewed literature

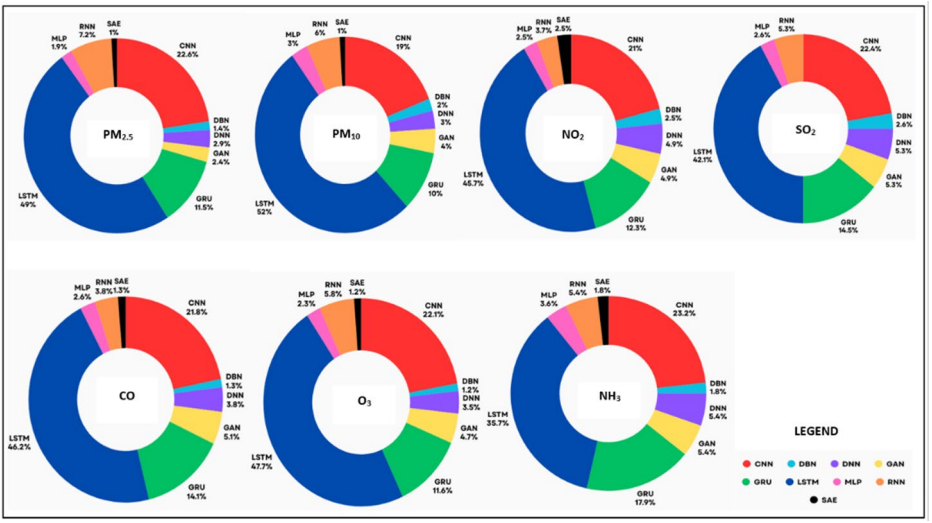


Fig. 5 Doughnut chart illustrating the distribution of predicted pollutants across all reviewed Deep Learning Families

6 Spatiotemporal deep learning architectures for air quality prediction

Air quality varies across both space and time, shaped by the interplay of emission sources, meteorological conditions, atmospheric transport, and regional characteristics. To model these complex dynamics effectively, DL approaches must be able to capture how pollutant concentrations evolve over time while also accounting for spatial relationships between monitoring locations. In response to this challenge, recent research has increasingly adopted graph-based spatial representations, combined with specialized temporal encoders and integration strategies, to learn from heterogeneous data sources and represent spatiotemporal dependencies in a unified modeling framework.

6.1 Spatial graph construction in deep learning models

In spatiotemporal DL, the spatial relationships between monitoring sites are commonly encoded through graph structures that describe how strongly one location interacts with another. Studies adopt several strategies to build these adjacency matrices, each reflecting different aspects of pollutant movement and atmospheric processes.

6.1.1 Static graphs

Static graphs use a fixed spatial structure throughout model training. These graphs are usually constructed from simple and stable relationships such as geographic distance, k-nearest-neighbor (kNN) connections, shared administrative boundaries, or long-term pollutant correlation patterns between monitoring stations. While distance and kNN-based graphs reflect direct physical proximity, correlation-driven graphs highlight stations that tend to behave similarly because of comparable emission sources or common meteorological influences. Several of the reviewed DL models adopt predefined static adjacency matrices, including GCN-ConvLSTM (Muthukumar et al. 2022a), AGCTCN (Choudhury et al. 2022), STHTNN (Chen et al. 2024b), KSC-ConvLSTM (Chen et al. 2025a), and static adjacency variants of DCRNN (Wang et al. 2022a). These models rely on fixed graph structures derived from spatial distance, neighborhood topology, or persistent pollutant correlations to encode stable spatial dependencies.

6.1.2 Dynamic graphs

Dynamic graphs continually update their structure to capture how pollutant transport pathways evolve over time. Unlike static graphs, the connections between monitoring locations are allowed to change in response to shifting dispersion patterns, wind behavior, or rapidly varying meteorological conditions. In many implementations, the diffusion or connectivity matrix is recalculated at each timestep, enabling the model to track non-stationary atmospheric processes and short-term variability. Among the reviewed models incorporate dynamic or time varying graph representations. DCRNN (Wang et al. 2022a), regenerates their diffusion matrices at every step to reflect the evolving transport dynamics. Models such as AC_GRU (Haghbayan et al. 2024), GNN-SAGE-QNN (Oliveira Santos et al. 2025), and VMD-GAT-BiLSTM (Wang et al. 2024b) learn adjacency structures that adjust automati-

cally to changing pollution correlation. Other spatiotemporal architectures, including ST-Conv, STENN, and STF-Net, utilize dynamically evolving graph parameters to represent short-term dispersion shifts and non-stationary atmospheric behaviors.

6.1.3 Learned or adaptive graphs

A growing class of spatiotemporal models moves beyond predefined spatial structures by learning the adjacency relationships directly from data. Instead of depending on distance-based or correlation-based graphs, these approaches infer spatial dependencies that may be hidden or nonlinear. Several reviewed articles like, AAMGCRN (Chen 2024), AHGCNN (Hu et al. 2024), and SpAttRNN (Huang et al. 2021) Use attention mechanisms or embedding-based strategies to compute context-dependent edge weights. These architectures dynamically adjust connectivity based on current input patterns, enabling the model to identify influential relationships that may shift over time. GEDE (Vu et al. 2024) and GNN-SAGE-QNN (Oliveira Santos et al. 2025) also infer latent spatial structures through graph embeddings, capturing subtle inter-station dependencies that are not apparent from physical distance alone. By uncovering hidden spatial influences, these learned graph approaches enhance predictive capability, particularly in heterogeneous environments.

6.1.4 Exogenous and multi-view graphs

Exogenous and multi-view graphs incorporate additional environmental and contextual factors to enrich spatial representation beyond simple geographic proximity. Instead of defining adjacency purely through distance or correlation, these graphs integrate variables such as prevailing wind direction and speed, terrain elevation, land-use type, traffic density, and emission-source distributions. Several reviewed models adopt exogenous-enhanced or multi-view spatial structures. AAMGCRN (Chen (2024) integrates multiple graph views. ST-AQF (Prado-Rujas et al. 2024), STF-NET (Zhang and Gan 2023), STN (Zhang and Zhang 2023), and STHTNN (Chen et al. 2024b) incorporate meteorology-augmented adjacency information and structural priors reflecting environmental drivers of pollutant transport. VMD-GAT-BiLSTM (Wang et al. 2024b) and AC_GRU (Haghbayan et al. 2024) leverage multi-factor attention, enabling the spatial graph to incorporate exogenous influences, while GEDE (Vu et al. 2024) uses graph embedding diffusion enriched by external factors. These models combine several spatial perspectives to achieve robust representation of complex atmospheric processes.

6.1.5 Edge weighting strategies

Edges within spatial graphs can be assigned weights in a number of ways, and this affects the way information is exchanged between nodes. Common strategies are inverse distance functions, gaussian similarity kernels, pollutant correlation score, attention derived coefficients or fully learnable parameters embedded into the model. These weighting choices have direct impacts on the smoothness and stability of message passing and ultimately the ways in which a model works to capture the pollutant transport pathways, cross-regional dependencies and complex, irregular spatial patterns that are characteristic of real-world air quality systems. Many reviewed GNN based models such as AAMGCRN (Chen 2024),

SpAttRNN (Huang et al. 2021), and AHGCNN (Hu et al. 2024) make use of attention based or learned weighting mechanism that allow to flexibly and context dependently modulate the edge. Others, such as GCN-ConvLSTM (Chen et al. 2024b), STHTNN (Chen et al. 2024b), and distance-based variants of DCRNN (Wang et al. 2022a), use predefined or physics informed weighting schemes based on geographical distance or pollutant similarity.

6.2 Temporal encoders for sequential pollutant dynamics

Temporal encoders dictate how models perceive the change of pollutant concentration over time, including characteristic activities like diurnal and weekly cycles, lagged effects from emission and more general seasonal variations. These encoders are basic to the modeling of non-stationary pollutant dynamics and have a decisive role in how well spatial and temporal information is fused downstream.

6.2.1 Gated recurrent networks

Recurrent architectures remain the backbone of temporal modeling in air quality forecasting. LSTM and GRU networks are widely used because their gating mechanisms can capture long-range dependencies and mitigate issues such as vanishing gradients, some of the examples in air quality forecasting include, BiLSTM (Kim et al. 2023), used to integrate forward and backward temporal context. Stacked-LSTM (Tello-Leal et al. 2024) and Lag-LSTM for deep, lag-aware temporal modeling. GRU-ED (Shakya et al. 2023) for multi-step forecasting. CNN-LSTM (Yafouz et al. 2021; Wardana et al. 2021; Bekkar et al. 2021; Uluocak et al. 2025), CNN-GRU (Yeo et al. 2021), and LSTM-GRU (Wang et al. 2019) hybrids where RNNs serve as the temporal encoder. These models consistently perform well in predicting $PM_{2.5}$, O_3, and NO_2 owing to their robustness against noise and non-stationary fluctuations. Hybrids such as CNN-LSTM (Lee et al. 2025), and CNN-GRU (Rad et al. 2025) utilize CNN layers for spatial feature extraction before temporal encoding with RNNs.

6.2.2 Temporal convolutional networks

Temporal Convolutional Networks (TCNs) use dilated causal convolutions to achieve large temporal receptive fields at a fairly low computational cost. This makes them good to model long sequences or high frequency datasets. Some of the examples include AGCTCN (Choudhury et al. 2022), STL-TCN-BiLSTM-DMAttention (Li and Jiang 2023), TCN (Wang et al. 2024a), TCN-SENet-BiGRU-CBAM (Mao et al. 2025). These models are good in modeling long sequence or high frequency datasets and have performed well in medium and long horizon forecasting tasks, particularly under rapid meteorological transitions.

6.2.3 ConvLSTM and convolutional RNNs

ConvLSTM extends LSTM by embedding convolutional filters within its recurrent gates, enabling simultaneous learning of spatial patterns and temporal evolution. Because of this dual capability, ConvLSTM based models are especially effective in gridded or raster-based inputs such as satellite AOD, CMAQ outputs, or meteorological fields. Models under this category include, ResNet-ConvLSTM (Zhang et al. 2022b), KSC-ConvLSTM (Chen et al.

2025a), C^{2}LSTM (Dey 2024), REG-CLSTM (Manna and Anitha 2024), VMD-CSA-CNN-LSTM (Guo et al. 2024), ODMSCNN-LSTM (Dai et al. 2021), Conv1D-BiLSTM (Ayus et al. 2023), etc. These architectures excel in capturing fine-scale dispersion dynamics and short-term fluctuations driven by meteorological variability and gridded inputs such as satellite AOD or CMAQ outputs.

6.2.4 Transformers and attention-based encoders

Transformer-style encoders have rapidly gained traction due to their ability to model long-range temporal dependencies through attention. Their non-recurrent structure avoids the limitation of sequential processing and enables efficient learning across long sequences. Some of the examples include, Kalman-Attention-LSTM (Zhou et al. 2022), hybrid Attention-based Seq2Seq (Wang et al. 2022a), SpAttRNN (Huang et al. 2021), STL-TimesNet-iTransformer-CPO (Ma et al. 2025), CNN-BiLSTM-Attention (Xinxin 2025), LSTM with Attention (Karnati et al. 2025), MEMD-L SHADE-Transformer (Tang et al. 2025). These models integrate attention-based temporal encoding with RNNs or graph-based spatial learning and are particularly effective when pollutant dynamics shift with seasons, boundary-layer behavior, or abrupt meteorological episodes.

Overall, the choice of temporal encoder strongly influences how effectively a model can capture pollution evolution. RNNs excel at handling non-linear sequential signals, TCNs provide efficient long-range modeling, ConvLSTM unify spatial and temporal learning and Transformer-based encoders deliver state-of the-art performance for multi-step, long horizon forecasting. These temporal components form a critical backbone for subsequent spatial-temporal fusion in DL air-quality models.

6.3 Fusion mechanisms in Spatiotemporal deep learning models

To provide a clearer conceptual organization of spatiotemporal DL approaches for air quality forecasting, this section adopts a formal taxonomy of fusion architectures. Based on the order and tightness of spatial–temporal integration, existing models can be grouped into five architectural families: (i) spatial-first, temporal-next pipelines; (ii) temporal-first, spatial-next pipelines; (iii) unified spatiotemporal blocks; (iv) attention-based fusion mechanisms; and (v) multi-graph and multi-source fusion architectures. This taxonomy clarifies design assumptions, comparative strengths, and typical deployment scenarios across the literature.

Fusion mechanisms determine how spatial patterns and temporal dynamics are blended within an integrated spatiotemporal forecasting framework. They influence model adaptability across heterogeneous regions, seasonal regimes, and rapidly changing meteorological conditions. Different fusion strategies emphasize spatial structure, temporal evolution, or balanced joint learning depending on the model design. Table 5 summarizes this taxonomy and contrasts the key architectural assumptions, representative models, and deployment contexts associated with each spatiotemporal fusion family.

6.3.1 Spatial-first architectures

In this design, spatial features extracted from GNNs or CNNs are fed into temporal encoders such as LSTM, GRU, TCN, or Transformer modules. This pipeline reflects the assump-

Table 5 Taxonomy of Spatiotemporal fusion architectures for deep Learning–based air quality forecasting

Architecture family	Core design principle	Canonical examples (illustrative)	Strengths	Most appropriate use cases
Spatial-First Architectures (spatial-first, temporal-next pipelines)	Spatial dependencies are learned first (CNN/GNN), followed by temporal sequence modeling (LSTM/GRU/TCN/Transformer).	CNN–LSTM; GCN–ConvLSTM; Geo-LSTM	Effective at capturing spatial interactions before temporal evolution; intuitive pipeline.	Dense monitoring networks; strong spatial correlation; short- to medium-term forecasts.
Temporal-First Architectures (temporal-first, spatial-next pipelines)	Temporal patterns are encoded first, then spatial relationships are imposed on learned temporal features.	Attention-based Seq2Seq; SpAttRNN; LSTM–CVT	Emphasizes temporal dynamics and lag effects; flexible for irregular spatial layouts.	Locations with strong diurnal/seasonal cycles; sparse or weak spatial coupling.
Unified Spatiotemporal Architectures (unified spatiotemporal blocks)	Spatial and temporal dependencies are jointly modeled within a single integrated neural block.	ST-Conv; STENN; AAMGCRN	Tight spatiotemporal coupling; improved modeling of transport and dispersion processes.	Regional-scale forecasting; short-horizon predictions; high-frequency data.
Attention-Centric Spatiotemporal Architectures (attention-based fusion)	Attention mechanisms dynamically weight spatial locations, time steps, or feature modalities.	CNN–BiLSTM–Attention; Transformer-based hybrids	Improves interpretability; adapts to non-stationarity and changing conditions.	Non-stationary environments; heterogeneous meteorology; decision-support settings.
Multi-Graph/Multi-Source Spatiotemporal Architectures (multi-graph and multi-source fusion)	Multiple spatial graphs and/or heterogeneous data sources are fused via gated or hierarchical mechanisms.	MTGNN; MG-AGCRN; DCRNN variants	Captures complex transport pathways and multi-scale spatial heterogeneity.	Large heterogeneous regions; multimodal data; cross-city or cross-network generalization.

tion that spatial interactions between monitoring stations or grid cells shape the subsequent temporal evolution of pollutants. Few examples of models using this strategy include GCN-ConvLSTM (Muthukumar et al. 2022b), CNN-LSTM (Mohapatra et al. 2025), ODMSCNN-LSTM (Dai et al. 2021), TL-BLSTM (Ma et al. 2019a), Geo-LSTM (Ma et al. 2019b), C-LSTME (Wen et al. 2019), AC-LSTM (Li et al. 2020). These models first capture spatial dependencies through convolution or graph layers and then encode temporal sequences through recurrent or attention-based modules.

6.3.2 Temporal-first architectures

Here, temporal features are learned first through RNNs, TCNs, or Transformer encoders. Spatial dependencies are then applied over temporally encoded features via GNNs or spatial attention layers. This strategy suits scenarios where temporal patterns (diurnal cycles, lag effects) are stronger or more stable than spatial correlations. Models falling under this category are Kalman-Attention-LSTM (Zhou et al. 2022), hybrid Attention-based Seq2Seq (Wang et al. 2022a), SpAttRNN (Huang et al. 2021), STL-TimesNet-iTransformer-CPO (Ma et al. 2025), LSTM-CVT (González-Enrique et al. 2021), CNN-LSTM-SE (Yang et al. 2024b), CNN-GRU (Yeo et al. 2021), BiAGRU (Zhang et al. 2020b), ADGRU (Lu et al. 2023), NLSTM (Liang et al. 2023). These architectures emphasize capturing sequential pollutant dynamics before imposing spatial constraints or graph-based reasoning.

6.3.3 Unified Spatiotemporal architectures

These architectures combine spatial graph convolution and temporal gating within the same neural block, enabling tightly coupled spatiotemporal learning without separate modules. These models are widely used in state-of-the-art traffic, climate, and air quality forecasting tasks. Models falling under this category STENN (Wang et al. 2022b), AAMGCRN (Chen 2024), ST-Conv (Chen et al. 2024a). By integrating spatial and temporal modeling in a single module, these models efficiently capture cross-regional pollutant transport and short-term dispersion patterns.

6.3.4 Attention- centric Spatiotemporal architectures

Attention mechanisms strengthen the model's ability to identify key regions, important timesteps, and dominant meteorological drivers. Attention can operate in spatial, temporal or cross-modal dimensions. Few examples are CNN-BiLSTM-Attention, (Xinxin 2025) LSTM with Attention (Karnati et al. 2025), MEMD-L SHADE-Transformer (Tang et al. 2025), WBiLSTM (Rezaali et al. 2024), TG-Hybrid (Wang et al. 2025). These architectures allow the network to dynamically weigh the importance of locations, past timesteps, or external features such as emissions and meteorology.

6.3.5 Multi-graph/multi-source Spatiotemporal architectures

Some models integrate multiple spatial graphs such as distance, road networks, meteorological similarity, or learned adjacency matrices or fuse multiple modal inputs. Fusion may occur through weighted averaging, gated fusion, hierarchical attention, or multi-branch encoders.

Models falling under this category include MTGNN, PMDNN, MG-AGCRN, Multi-graph GCN, DCRNN variants with exogenous inputs, ST-MetaNet. These mechanisms improve the model's ability to capture pollutant transport across heterogeneous regions.

6.4 Evidence from ablation studies and geographic domains

A large portion of recent spatiotemporal DL studies include ablation experiments to isolate the effect of individual components like spatial modules, temporal encoders, attention layers, or learned graph structures. Evidence from these studies shows how much each component contributes to overall predictive performance.

6.4.1 Impact of Spatial modules

Removing the spatial graph module and comparing against temporal-only baselines such as LSTMRNNs, P-LSTM (Aggarwal and Toshniwal 2021), RNN-LSTM (Arsov et al. 2021), Bi-LSTM (Kim et al. 2023), Geo-LSTM (Ma et al. 2019b), etc., typically leads to poorer RMSE and R^2 performance. This highlights the importance of explicitly modeling spatial dependencies between monitoring station grid cells, as spatial-only or temporal-only models fail to capture cross-regional pollutant transport.

6.4.2 Role of attention layers

Dropping attention components, for instance in Kalman-Attention-LSTM (Zhou et al. 2022), hybrid Attention-based Seq2Seq (Wang et al. 2022a), SpAttRNN (Huang et al. 2021), CNN-BiLSTM-Attention (Xinxin 2025), often reduces model accuracy and reveals that attention mechanisms help prioritize relevant features. Attention mitigates over smoothing in densely connected graphs and improves the interpretability of key spatial or temporal drivers.

6.4.3 Effect of learned or adaptive graphs

Replacing static adjacency with adaptive or learned graphs generally improves performance, as demonstrated in AGCTCN (Choudhury et al. 2022), AAMGCRN (Chen 2024), DCRNN (Wang et al. 2022a), VMD-GAT-BiLSTM (Wang et al. 2024b). This confirms that pollutant transport pathways are context-dependent and can vary with weather, topography, or emissions, which static graphs fail to capture.

6.4.4 Temporal encoder evaluation

Testing different temporal encoders shows that models with stronger long-range sequence modeling abilities, such as Transformer-based models (Ma et al. 2025; Tang et al. 2025), consistently perform better on long-horizon forecasts compared to vanilla LSTMs or GRUs. This emphasizes the importance of encoder choice in capturing diurnal cycles, lagged emissions, and seasonal variability.

6.4.5 Single-graph vs. multi-graph fusion

Comparing single-graph fusion models to multi-graph or multi-source approaches highlights the added value of incorporating meteorological, emission, or terrain-related information. Multi-graph fusion improves robustness, stability, and generalizability across heterogeneous geographic domains. Where these ablation experiments are missing, studies frequently acknowledge that the absence of component-level evaluation restricts causal interpretation and limits understanding of which modules drive predictive success.

Spatiotemporal studies are recorded based on whether models incorporated graph-based spatial representations, temporal encoders, and hybrid or fusion mechanisms. When reported, details such as adjacency construction, use of static or dynamic graphs, edge-weighting strategies, and the presence of ablation analyses were also noted and are illustrated in Table 6. Because most studies provided only high-level methodological descriptions, this categorization is descriptive rather than causal, but it allows a structured synthesis of spatio-temporal design patterns across the reviewed literature.

7 Comprehensive evaluation of DL models beyond evaluation metrics

Most DL studies on air quality forecasting mostly report accuracy measures such as RMSE, MAE, R^2. While these are measures of the prediction error, they capture only part of model performance. Important aspects such as robustness, generalization, computational efficiency and external validity tend to be overlooked. This section summarizes these more general evaluation dimensions.

7.1 Robustness and hyperparameter sensitivity

DL models are very sensitive to architectural design choices and hyperparameter configurations. However, only a few studies perform a systematic robustness analysis or sensitivity evaluation. Even slight changes in dropout learning rates, or the number of hidden units, have a great impact on model stability, especially in cases where training data is noisy or has missing values. Model robustness is also very closely related to data quality. Studies based on poorly standardized inputs or unbalanced datasets often have a higher variability of performance, regardless of careful hyperparameter tuning. These observations highlight the importance of consistent preprocessing pipelines and the use of standardized benchmark datasets for comparing DL models for air quality forecasting.

7.2 External validity and generalizability across contexts

DL models tend to have problems when deployed in different regions than those used to train them. Models that have been developed in urban or temperate conditions tend to suffer degraded performance when used in the tropical, coastal, industrial, or rural environments due to differences in climate, emission sources, and meteorological conditions as well as land-use patterns. Although recent attempts have been made to investigate ways such as transfer learning, domain adaptation and multi-site validation to improve external validity, generalizability is one of the most important and least understood challenges. Evi-

Table 6 Key characteristics of Spatiotemporal deep learning models: graph design, Temporal encoders, fusion schemes, and ablation support

Model	Spatial Graph	Adjacency Construction	Temporal Encoder	Fusion/Hybrid Mechanism	Edge Weighting	Ablation Study	Reference
DCRNN	Directed diffusion Graph	Predefined via distance and diffusion	GRU	Diffusion Convolution + GRU	Learned diffusion weights	No	(Wang et al. 2022a)
GT-LSTM	Learned Adaptive Graph	Adaptive Adjacency matrix	LSTM	Graph + LSTM hybrid	Learned	Yes	(Mao et al. 2021)
GCN-ConvLSTM	Predefined - Physical Adjacency Graph	Distance based Adjacency	ConvLSTM	GCN + ConvLSTM	Binary	No	(Muthukumar et al. 2022a)
SpAttRNN	Attention-based dynamic Graph	Learned via attention	RNN	Graph attention + RNN	Learned via attention scores	Yes	(Huang et al. 2021)
AGCTCN	Adaptive Graph	Adaptive Adjacency	TCN	Graph + TCN	Learned	Yes	(Choudhury et al. 2022)
CNN_GRU	Predefined Adjacency Graph with Attention	Distance based Adjacency	GRU	Graph Convolution + GRU	Learned via attention	No	(Haghbayan et al. 2024)
AHGCNN	Hierarchical multi-scale Graph	Multi-Scale adjacency	CNN/Graph convolution	Hierarchical fusion	Learned per level	Yes	(Hu et al. 2024)
AAMGCRN	Multi-graph	Multiple Adjacency Matrices	GRU	Multi- Graph fusion + GRU	Learned per graph	No	(Chen 2024)
STHTNN	Dynamic spatiotemporal graph	Learned from data	RNN/Temporal Attention	ST graph + temporal fusion	Learned	Yes	(Chen et al. 2024b)
VMD–GAT–BiLSTM	Learned graph via GAT	Adaptive Adjacency via attention	BiLSTM	Graph attention + BiLSTM	Learned via GAT	No	(Wang et al. 2024b)
GEDE	Learned adjacency graph	Data-driven adjacency	GRU	Graph + GRU hybrid	Learned Dynamically	Yes	(Vu et al. 2024)
MSTGAN	Initial distance-based adjacency + learned STDG	Initial adjacency via distance threshold + STDG learned via bilinear attention	Multi station Transformer + GRU	Multi-scale ST blocks + 2D Conv fusion	Initial adjacency + learned STDG	Yes	(Zhou et al. 2024)
GNN-SAGE-QNN	GraphSAGE - based graph	GraphSAGE aggregation	QNN	Graph embedding + QNN	Learned via node aggregation	No	(Oliveira Santos et al. 2025)
KL-PV-CBGRU	Predefined adjacency + PV correlation graph	Distance based + correlation	CBGRU	GCN + BiGRU hybrid	Learned via CBGRU	Yes	(Chen et al. 2025b)

dence from the reviewed literature suggests three main forms of generalization: geographical transfer, seasonal transfer and station-level generalization. Across these situations, the model performance tends to drop by around 15–45% when they are tested under conditions that deviate from that of the training data, and the need to expose them to more robust and consistent validation protocols.

7.2.1 Geographical transfer

To assess portability across cities or regions, several studies use leave-city-out evaluation, training on multiple cities while testing on an unseen one. These analyses consistently show that DL models often overfit to the pollutant-meteorology relationships of the training area. When exposed to a new region with different emission profiles, climate patterns, or terrain, performance drops significantly. Domain adaptation strategies such as adversarial alignment, fine tuning, or transfer learning can improve transferability, but they still rely on having at least some data from the target region. Thus, fully region-invariant generalization remains an open challenge.

7.2.2 Seasonality transfer

Seasonal generalization is similarly limited. Models calibrated on winter conditions that are dominated by inversion events, biomass burning, calm winds and stagnation frequently perform poorly during summer or monsoon periods, where photochemistry and dispersion behave differently. Only a small number of studies adopt seasonal cross-validation or season-specific train-test splits, and these consistently show sensitivity to humidity, temperature, wind variability, and photochemical processes. Emerging solutions such as periodic encodings, seasonal embeddings, or cycle-aware architectures show early promise in reducing seasonal drift, but the field still lacks comprehensive seasonal generalization benchmarks.

7.2.3 Station-level generalization

Generalization challenges are also faced within the same city from different monitoring stations. Many DL models have the tendency to implicitly memorize station-specific noise and not learn spatial representations that are actually transferable. Evidence from multi-site cross validation (where each monitoring station is held out in turn) often shows significant performance degradation (‘not robust’ with respect to intra-city variation). The recent advances in architectural designs like GCN and GC-LSTM hybrids, attention-based spatial fusion mechanisms, periodic encodings and cycle-aware modules have illustrated improvement by explicitly modeling over spatial and temporal dependencies. While these designs improve the stability, they do not entirely solve the overfitting at the station level.

7.2.4 Criteria for robust versus fragile generalization

Overall, the reviewed literature shows that DL models have a low level of external validity unless backed by rigorous evaluation protocols. Studies using leave-city-out validation, multi-site cross-validation, or transfer learning baselines are more reliable evaluations than using random data splits. To move closer to deployable air quality forecasting systems,

future research should focus on the use of multi-city datasets, explicit domain adaptation approaches and fixed generalization benchmarks.

Across the reviewed studies, robust generalization is consistently associated with evaluation protocols that involve spatially or temporally disjoint test sets, such as leave-city-out, multi-site, or season-held-out validation, as well as explicit modeling of spatial or temporal dependencies. In contrast, fragile generalization typically arises from single-city or single-station evaluations, random train-test splits, limited temporal coverage, and the absence of external validation. Distinguishing between these evaluation designs is essential for interpreting reported performance and assessing deployment readiness.

7.3 Overfitting and regularization

Overfitting is a constant issue with DL-based Air Quality Modeling, and it is mainly caused by the lack of availability of high-resolution spatiotemporal data. Regularization strategies such as dropout, early stopping, L2 penalties and cross-validation are recommended but are not used consistently across studies. In addition to architectural design, there are a number of methodological practices that contribute to overfitting. These include improper or undisclosed train-validation-test splits, which can cause overly optimistic estimates of performance when the test data have an unintended effect on model tuning. Random data partitioning for time series forecasting has the additional problem of introducing temporal leakage, for multi-step prediction horizons rolling origin or sliding window evaluation schemes are more suitable. Additional leakage may occur from satellite observations, reanalysis products or chemical transport model outputs if variables accidentally encode future information. Such inputs should be duly lagged and tuned to their release times. Model selection should be done with strict reliance on validation performance and the test set should be used for one final evaluation only. Nested validation strategies are even more reproducible and robust. Although some studies perform sensitivity analyses (for example, of the input windows, subsets of the pollutants, or of the spatial adjacency matrices), these evaluations are quite rare. Wider adoption would yield better knowledge on model stability and address the overfitting risks.

7.4 Computational efficiency

As DL architectures become more complex and deeper, training and inference costs are significantly increased. While such models may provide better predictive accuracy, they may not be practical for real-time forecasting or for deployment in resource-constrained environments. Simpler alternatives, such as lightweight CNNs, shallow recurrent networks, and small transformer versions, provide useful trade-offs between computational efficiency and performance. Consequently, computational complexity should be taken together with predictive accuracy when models for operational air quality forecasting systems are selected.

7.5 Standardized performance metrics for model comparison

In order to ensure consistency and fairness across the studies, frequently used evaluation metrics were defined using unified notation. Table 7 shows the mathematical expressions of RMSE, MAE, MAPE, R^2, NSE and NRMSE. Units were standardized and forecast horizons

were created in similar groups. Pollutant concentrations that were originally reported in different units (e.g. mg/m^3) were converted to ug/m^3 for consistency. Because several studies reported missing summary statistics (e.g. mean, min, max values of pollutants), NRMSE could only be calculated if there was enough information available. In cases where this was not possible the reported metrics were kept and explicitly flagged as "Not Reported (NR)" for NRMSE, in order to maintain transparency. Additional harmonization steps, such as concentration unit and averaging interval, were added to ensure comparability across the reviewed literature. A consolidated summary of these standardized metrics is given in Supplementary Table ST5.

7.6 Structured quantitative synthesis

To contrast a variety of forecasting approaches used in the literature, the evidence was organised in a three-stage synthesis. The first stage, which is presented in Supplementary Table ST6, collates study level technical information including the study, setting, modeling setup, pollutant type, forecast horizon and originally reported performance metrics. Before proceeding further, eligible studies, i.e. those for which normalized error metrics could be computed, were identified and screened. The second stage, which is summarized in Supplementary Table ST7, is used to standardize all eligible studies into one consistent evaluation format. This step is to ensure that the differences in the definition of metrics, train-test splits and evaluation strategies can be interpreted in a consistent way across different modeling frameworks. The last stage, shown in Supplementary Table ST8, is the aggregation of the overall ranking that was achieved by the considered models. It summarizes evidence based on average ranks, variability ranges and number of studies in which each model occurs. The distribution of NRMSE values for the pollutants is presented in Supplementary Table ST9 and Fig. 6. Together, these three stages provide a definite progression - from raw study characteristics to harmonized performance metrics to consolidated model-level insights. This structured process makes it possible to have a more transparent and balanced understanding of how various DL approaches work across pollutants, regions and forecasting tasks.

Despite the structured harmonization strategy, substantial heterogeneity persists across the reviewed studies. This heterogeneity spans multiple dimensions, including geographical context, pollutant targets, temporal resolution, forecasting horizons, dataset duration, feature sets, and validation protocols. In addition, the literature exhibits considerable variability in reported performance metrics, baseline definitions, and experimental setups. Under such conditions, pooled effect-size-meta-analysis is statistically inappropriate and risks producing misleading inferences due to violations of comparability and independence assumptions.

Consequently, quantitative aggregation was restricted to cases where a minimum level of conceptual alignment was present, such as comparable pollutants, forecasting horizons, and normalized error metrics. When these conditions were not satisfied, aggregation was deliberately avoided and synthesis focused instead on relative performance patterns and robustness trends across model families. Median-based summaries and rank-based comparisons were therefore adopted as the primary synthesis tools, as they are robust to extreme values, less sensitive to distributional assumptions, and remain interpretable under heterogeneous reporting practices. This approach is consistent with evidence synthesis recommendations for rapidly evolving ML domains where methodological standardization is still limited.

Table 7 Mathematical definitions of standard performance metrics

Metric	Full Name	Mathematical Expression
R^2	Coefficient of Determination	$1 - \frac{\sum_{i=1}^{n}(Y_i - \widehat{Y_i})^2}{\sum_{i=1}^{n}(Y_i - \overline{Y_i})^2}$
MAE	Mean Absolute Error	$\frac{1}{n}\sum_{i=1}^{n} \mid Y_i - \widehat{Y_i} \mid$
RMSE	Root Mean Squared Error	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{Y_i})^2}$
NSE	Nash-Sutcliffe Efficiency	$1 - \frac{\sum_{i=1}^{n}(Y_i - \widehat{Y_i})^2}{\sum_{i=1}^{n}(Y_i - \overline{Y_i})^2}$
MAPE	Mean Absolute Percentage Error	$\frac{100}{n}\sum_{i=1}^{n} \mid \frac{Y_i - \widehat{Y_i}}{Y_i} \mid$
NRMSE	Normalized RMSE	$\frac{RMSE}{Y_{max} - Y_{min}}$

Where, Y_i = observed concentration, $\widehat{Y_i}$ = predicted concentration, n = number of samples, and $\overline{Y}$ = mean concentration

Beyond aggregation of numbers, we performed a critical review of the methodological heterogeneity of the reviewed studies. Variations in the design of the cross validation, in tuning strategies for hyperparameter values, and input feature selection were shown to have a large effect on reported performance, even for the same pollutants and forecast horizons. In particular, studies with multi-site cross validation protocols or leave-city-out evaluation protocols exhibited a higher level of robustness while the use of hold-out splits or single-site validation were frequently associated with overoptimistic performance estimates. By jointly examining these methodological choices in conjunction with quantitative metrics of performance, this review represents a more complex interpretation of the usefulness of these models, and underscores both methodological strengths and potential sources of bias in the current literature. Since the reported metrics were not consistent across studies, only articles which fulfilled the eligibility criteria (e.g., reporting NRMSE values) were used in the quantitative evaluation. To overcome this limitation, we propose a standardized evaluation framework for benchmarking as shown in Table 8. This framework contains a list of essential metrics (RMSE, MAE, R^2 and correlation coefficients) for assessing the predictive accuracy and comparison with observations as well as normalized metrics (sMAPE, MASE) for fair comparison between different regions or pollutants. Consistent forecast horizons and well-defined splits between training and testing data help to preserve the temporal dependencies and avoid data leakage. Assessing performance in urban, suburban, and rural locations helps to establish the applicability of the model in a variety of situations. Transparent reporting of both absolute and normalized errors further increases reproducibility, facilitates fair benchmarking, and allows for evidence-based conclusions across the literature.

7.7 Reproducibility, transparency and minimum reporting requirements

Across the reviewed literature, recurrent issues related to reproducibility and transparency were identified, frequently limiting the interpretability, comparability, and reuse of published results. Common shortcomings included incomplete descriptions of data preprocessing pipelines, unclear train–test splitting strategies, insufficient reporting of hyperparameter configurations, and inconsistent disclosure of evaluation protocols. These limitations complicate replication efforts and hinder fair benchmarking across studies, particularly in a field characterized by heterogeneous datasets, forecasting horizons, and modeling assumptions.

To overcome some of these challenges, in this review, we will suggest a minimal or 'core list' of reporting items specifically geared towards DL-based air quality forecasting studies. Rather than an uncompromising protocol, these items are designed as a sensible checklist that represents recurring gaps noticed throughout the surveyed literature. The goal is to make a small reference guide available for authors, reviewers and practitioners that will promote reproducibility, methodological clarity and transparent comparison of model performance.

Table 9 summarises the proposed minimum reporting checklist, including aspects of data provenance, spatial and temporal resolution of the data, specification of the model architecture, the training and validation procedures, metrics for evaluating the surrogate model every day, and computational aspects. The checklist focuses on things that were reported least in previous studies and that have a direct impact on the interpretation of the results and reproducibility. By placing these items in one place, the table allows the readers to easily

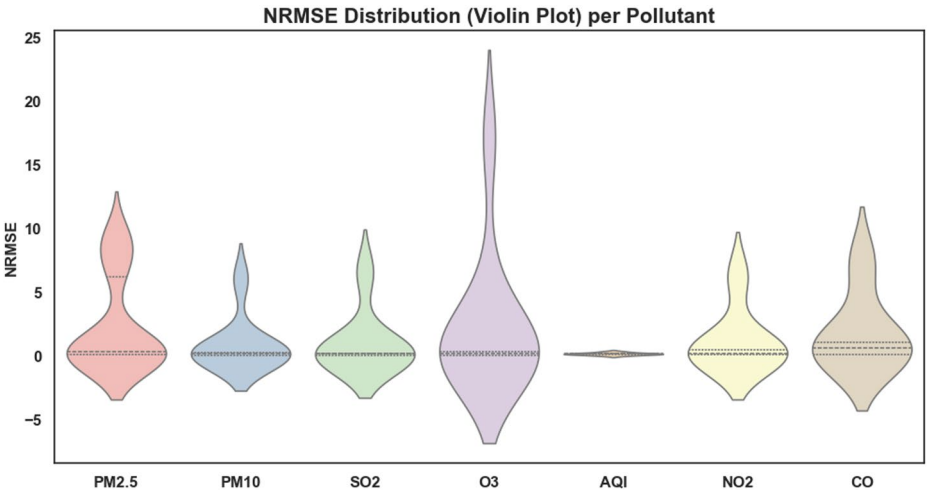


Fig. 6 Distributional patterns of normalized pollutant concentrations visualized using violin plots to highlight spread and density

Table 8 Standardized evaluation framework for deep Learning–based air quality forecasting

Component	Recommended Practice	Purpose/Benefit
Primary Metrics	RMSE, MAE, R^2/correlation coefficient	Quantifies prediction errors and agreement with observed values; widely recognized and interpretable
Complementary Metrics	sMAPE, MASE (optional)	Normalizes errors for fair comparison across pollutants, regions, and scales
Forecast Horizon	Short-term (1–24 h), multi-step (24–72 h)	Maintains consistency in temporal evaluation and enables benchmarking across studies
Data Splitting	Chronological or rolling-origin split	Preserves temporal dependencies and prevents data leakage
Spatial Evaluation	Evaluate separately for urban, suburban, and rural locations	Accounts for spatial heterogeneity in model performance
Reporting Guidelines	Report absolute and normalized errors, clearly define metrics	Improves reproducibility, transparency, and cross-study comparability

and quickly gauge the completeness of the reporting as well as support more consistent documentation practices in future work.

Adoption of these minimum reporting items would greatly enhance transparency and comparability across studies and will support cumulative evidence synthesis, as well as reduce ambiguity in claims of reported performance. More broadly, more transparent reporting standards can help bring about quicker transfer of DL models from exploratory research to operational air quality forecasting systems, where reliability, accountability and reproducibility are critical.

7.8 Common pitfalls and misconceptions

Despite the rapid improvement in methodology, the reviewed literature shows a number of recurring pitfalls and misconceptions that can lead to overoptimistic claims of performance,

Table 9 Minimum reporting checklist for deep Learning–based air quality forecasting studies

Reporting Domain	Minimum Reporting Items
Data and study context	Study region(s) and monitoring network; temporal coverage; pollutants modeled; data sources (ground, satellite, reanalysis); handling of missing data.
Input features and preprocessing	Meteorological variables used; lag structure and input windows; normalization or scaling methods; feature selection strategy.
Model architecture	Complete description of model components; spatial graph or adjacency definition (if applicable); rationale for architectural choices.
Training configuration	Loss function; optimizer; learning rate schedule; batch size; number of epochs; regularization methods.
Validation and evaluation protocol	Train–validation–test split strategy; temporal ordering; cross-validation design (e.g., multi-site or leave-city-out); avoidance of data leakage.
Performance metrics	Absolute error metrics (e.g., RMSE, MAE); normalized metrics (e.g., NRMSE, sMAPE, MASE); forecast horizons evaluated.
Baselines and comparisons	Definition of baseline models; fairness of comparisons; consistency of evaluation settings across models.
Generalization assessment	Evidence of geographical, seasonal, or station-level transfer; description of robustness or sensitivity analyses.
Computational aspects	Hardware used; training time; inference latency; model size or parameter count.
Reproducibility artifacts	Availability of code, trained models, or configuration files; data accessibility; random seed control.

low replication ability, and poor deployment readiness. These problems can result from the model design itself or from decisions on evaluation, data handling practices and some implicit assumptions regarding generalizability and operational feasibility. Identifying these pitfalls is fundamental for the proper interpretation of any reported result and for more robust and deployable systems for future research.

One of the common misconceptions is that the improvements inaccuracy metrics compared to conventional models are equivalent for real modeling progress. Many studies have found marginal improvements in RMSE or MAE but have not examined if these improvements are maintained by other validation protocols, locations, or over time. As was discussed in Sect. 7.1 and Sect. 7.2, it's a well-known fact that models that perform well when splitting the data randomly (train-test splits) usually show large advantages when the evaluation set are spatially or temporally disconnected from the sections the model has access to (which is a small subset of the whole) - these models are prone to fragile generalization. Without rigorous robustness analyses and validations of prediction measures from independent validators, reported gains can represent dataset-specific artifacts versus generalizable predictive capability.

A second common error is failure to properly address generalization and external validity. Often models have been trained and validated using single-city or single-station data sets, implicitly assuming that learned pollutant-meteorology relationships will be valid in other regions and/or seasons or at other monitoring sites. Evidence summarised in Sect. 7.2 shows that often this assumption does not hold in practice, with performance drops of 15–45% being frequently observed under geographical, seasonal or station level transfer. Failure to recognize these limitations may result in unrealistic expectations as to deployment readiness.

Overfitting and data leakage are another very important and often underestimated problem. As discussed in detail in Sect. 7.3, inappropriate train-validation-test divisions, leakage

of information in time series forecasting, and no care with future information in auxiliary data sources can exacerbate reports of inflation of performance. In the absence of transparent reporting of splitting strategies as well as preprocessing pipelines, doubts regarding the credibility of published results may exist, and reproducibility of the results by others may not be possible.

Computational efficiency as well as system-level feasibility are, usually ignored. While more and more complex architectures can have improved offline accuracy, are often accompanied by prohibitive training and inference cost. As described in Sect. 7.4 and 8.5, where the model neglects latency situations, memory footprint and operational limitations, it may lack control for reliable prediction an early warning applications. Treating predictive accuracy as the only measure of model quality therefore constitutes a limited and perhaps misleading way of evaluating a model.

Finally, there is an increasingly common (but as yet, insufficiently addressed) misconception that explainability and ethics are optional post hoc add-ons. As it was stressed in Sect. 8.4, these factors-married to the problem of lack of interpretability, absence of how to communicate uncertainty and assess fairness-could undermine trust, regulatory usability and fair deployment, despite a high level of predictive performance. Models that do not take into account uneven sensor coverage, as well as demographic differences may inadvertently perpetuate environmental inequalities.

Table 10 summarizes these common pitfalls, their typical manifestations in the literature, their practical consequences, and recommended mitigation strategies grounded in the evaluation and deployment principles discussed throughout Sects. 7 and 8. By explicitly recognizing these issues, future studies can move beyond incremental accuracy improvements toward more reliable, transparent, and operationally meaningful air quality forecasting systems.

8 Practical challenges in real-world deployment

Although DL models have been shown to have a high ability to predict, applying them to real-world scenarios where air quality is concerned has its own specific practical challenges such as inconsistent data, high computational requirements, and lack of interpretability. Despite such barriers, a number of operational initiatives demonstrate the increasing value of DL in actionable air-quality management. For instance, the Beijing Municipal Environmental Monitoring Center uses LSTM-based and hybrid models to forecast $PM_{2.5}$ hourly to timely alert high pollutant episodes. South Korea's National Institute of Environmental Research applies CNN-LSTM architectures to combine satellite AOD and ground sensors to enhance predictions of nationwide AQI, especially in areas with sparse monitoring. For the European Union, the Copernicus Atmosphere Monitoring Service (CAMS) has been piloting DL-based data assimilation to improve the outputs of chemical transport models for regulatory forecasting. Additionally, smart-city deployments in Singapore and Barcelona are using IoT sensor networks with lightweight edge-AI models to allow for real-time monitoring on traffic corridors. Collectively, these examples show that DL systems are moving from research prototypes to operational tools that can help to support early warning alerts, exposure management, and evidence-based policy interventions.

Table 10 Common pitfalls and misconceptions in deep Learning-based air quality forecasting studies

Pitfall/Misconception	Typical manifestation in the literature	Practical consequence	Recommended mitigation
Accuracy gains imply robustness	Reliance on RMSE/MAE improvements under random train–test splits	Overoptimistic performance claims; poor transfer to new settings	Use spatially and temporally disjoint validation (e.g., leave-city-out, season-held-out); report robustness analyses
Implicit generalizability	Evaluation limited to single city, season, or station	Large performance degradation under real-world deployment	Explicitly test geographical, seasonal, and station-level generalization
Inadequate handling of overfitting	Limited or undisclosed regularization and validation protocols	Inflated accuracy; unstable predictions	Apply strict temporal splits, nested validation, and sensitivity analyses
Data leakage in time-series forecasting	Random splits or use of future-informed auxiliary variables	Artificially high predictive performance	Enforce chronological splitting and align inputs with real data availability
Ignoring computational constraints	Focus on complex architectures without reporting resource usage	Models unsuitable for real-time or edge deployment	Report inference time, memory footprint, and latency–accuracy trade-offs
Model-centric evaluation only	Exclusive focus on offline regression metrics	Poor end-to-end system performance	Complement accuracy metrics with system-level indicators (latency, throughput, robustness)
Explainability treated as optional	Black-box models without interpretation or uncertainty reporting	Reduced regulatory trust and usability	Integrate XAI, uncertainty quantification, and transparent reporting
Neglect of fairness and representativeness	Training biased toward well-monitored urban areas	Reinforcement of environmental inequities	Use multi-site validation, fairness diagnostics, and coverage-aware modeling

8.1 Data quality and availability

Effective DL training is highly dependent on continuous, high-resolution and well-calibrated datasets. In many areas of the globe, especially in resource-limited areas, monitoring networks are patchy, sensors may fail or go out of calibration, and missing observations are commonplace. These limitations make the models less reliable and limit its generalizability in different environments.

8.2 Computational demands

Sophisticated DL architectures, in particular hybrid and multi-branch models, are computationally expensive. Training and deploying these models often require the use of modern graphics processing units (GPUs), substantial memory capability and, in some cases, access to cloud-based infrastructure. Such requirements are not always possible for agencies or institutions with limited financial or technical resources.

8.3 Forecasting constraints

Models that work well in a controlled research setting may not work well in an operational setting, where the generation of predictions must be fast and efficient. Low latency forecast-

ing, which is crucial in early warning systems, requires small and optimized architectures. Techniques like model compression and progress in edge computing technologies helps bring real-time deployment to a practical level.

8.4 Explainability and ethical considerations

Making DL models for air quality prediction both interpretable and ethically deployed is of essential importance for their real-world adoption. Although these models can be highly predictive, the complexity of such models can hinder transparency and foster mistrust among policymakers and regulators as well as the general public. The application of post hoc explainable artificial intelligence (XAI) techniques, such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), is used to offer transparency, that is the possibility to interpret systematically the outputs of models (Inam et al. 2025a, b). XAI has thus become an integral part of DL-based air quality forecasting addressing important issues regarding model interpretability and decision transparency (Yang et al. 2022; Agbehadji and Obagbuwa 2025; Inam et al. 2025c).

It is also not enough that forecasts are accurate in order to be used in the context of regulatory action, action to protect the public's health and risk communication in the event of extreme pollution events. XAI approaches such as SHAP, LIME, attention mechanisms, saliency-based methods enable models to demonstrate the role played by pollutants, meteorological variables and spatiotemporal patterns in composing the predictions, and thereby exposing the underlying reasoning of complex DL systems. By highlighting driving factors of pollution spikes and symptoms of model instability, these techniques help model outputs achieve more confidence and contribute to informed policymaking.

Ethical and governance issues are just as important. Issues relating to data privacy, calibration bias of sensors and bias of algorithms have to be handled with care, more so for IoT-enabled monitoring environments. Establishment of governance structures that prioritize responsible data usage, reporting, and human oversight is a crucial steps in ensuring that data can be used in a trustworthy manner. In order to concretely highlight the importance of considering explainability, uncertainty and fairness in decision making in the final decision, the next subsections outline structured use cases where DL-based forecasting outputs can be operationalized in regulatory and public health contexts.

8.4.1 Explainability for regulatory decision-making

In the control of air quality in cities, the SHAP-based feature attribution can be used to detect the key sources of pollutants which are responsible for the concentration peaks, allowing regulators to focus on specific actions such as traffic restrictions or emission controls from industries. Recent studies on PM_{10} forecasting show that feature importance analyses can point out emission sources and meteorological factors that have the strongest influence on the concentration of pollutants predicted (Inam et al. 2025d). Moreover, several papers have been able to use SHAP and LIME to interpret DL-based AQI forecasting models, which improved the transparency and usability of the predictive outputs for use by regulatory persons (Mohan and Abraham 2024; Mohapatra et al. 2025). By directly relating the model predictions to actionable mitigation strategies helps increase confidence of the predictions in regulatory policies, making it easier to form evidence-based policies and to implement

regulatory measures on air quality management practices, all while ensuring accountability of those implementations.

8.4.2 Uncertainty quantification for public-health advisories

Uncertainty quantification plays an important role in the advice to the public based on air quality forecasts. When DL models offer prediction intervals, probabilistic exceedance estimates or confidence bounds, authorities can issue health warnings if concentrations of pollutants are forecast to exceed regulations with a high degree of certainty. This approach can help minimize unwanted public alarm, emergency response resources and improve timing and credibility of advisories regarding outdoor activity restrictions, school closures or vulnerable population guideline dependencies. Evidence from climate-driven infectious disease forecasting shows that open reporting of uncertainty is an effective way to increase stakeholder trust and willingness to follow public health recommendations, as it indicates the relevance of such approaches for situations in air quality decision making as well (Inam 2025). Incorporating uncertainty-aware forecasting therefore advances both the trust and acceptance of DL-based air quality early warning systems by society.

8.4.3 Fairness and ethical deployment across communities

Fairness considerations are needed to ensure that DL-based air quality forecasting systems impart fairness of protection to different neighborhoods and different demographic groups. Models designed to be chiefly informed by all available well-instrumented urban stations, for example, may thus systematically fail to predict pollution levels at peri-urban, rural or socioeconomically disadvantaged sites, reinforcing environmental inequalities. Inclusion of fairness metrics, multi-site validation and spatial bias diagnostics can help decision-makers to detect such divergence and address any differences in monitoring strategies or mitigation approaches. Importance of Fairness: Making sure that prediction works fairly is not only an ethical responsibility but is also prerequisite to reliable deployment of such systems in operation. Addressing fairness and representativeness helps to improve the social legitimacy of DL-based forecasts, aids in the achievement of environmental justice goals, and also makes AI-driven decision support systems for air quality management overall more robust.

8.5 Deployment considerations and resource management

Deploying DL models outside the research environment may impose many problems, including computing power, infrastructure constraints, and the requirement for real-time performance. Larger architectures like LSTMs, CNNs and hybrid versions of the same not only often require large amounts of GPU memory, which is not appropriate for mobile or IoT/Edge devices without modifying it. Practical deployment therefore includes estimating resource budgets, which are associated with processing time, energy use and memory usage. There are several optimization strategies that can be used to reduce model complexity: Pruning, where redundant neurons, filters or weights are eliminated to decrease the model size and make it faster to infer while keeping the performance loss within an acceptable range; Quantization, where the precision of the weights is decreased to reduce memory usage and computation required to make inferences while still keeping performances to an acceptable

level; Knowledge distillation, where a smaller model is trained to mimic the behaviour of a larger one, while retaining most of the original model's predictive power with a much lower latency and resource demands. Edge deployments also require fast inference pipelines and sensor drift and missing data robustness. Continuous monitoring performance followed by retraining and regular data quality checks are necessary to ensure the reliability of the project on a long-term basis.

While these considerations are related to feasibility in the computational setting, in the reviewed literature evaluation is largely model-centric, with focus on traditional regression measures such as RMSE, MAE, and correlation coefficients. These metrics are necessary to evaluate the predictive accuracy of a system, but are unable to capture the end-to-end operation in the course of a real-world system setting. In contrast, the system-level performance includes factors such as data ingestion latency, inference throughput, update frequency, resiliency to receiving late or missing inputs, and how quickly forecasts may be used for decision-makers to take regulatory action or to generate public health advisories. Only a limited subset of those studies report evidence on deployment-oriented issues, such as real-time inferences feasibility, suitability for streaming sensor data, or the trade-off between accuracy and latency under edge-based or cloud-based deployment restrictions. As a result, models that show good offline predictive performance may still not be suitable for practical use because of excessive latency to provide a prediction, instability under the actual data examined in the original dataset or because of resource limitations. This review therefore highlights the need to understand clearly and differentiate predictive accuracy from the end-to-end system performance. Future air quality forecasting studies are encouraged to complement conventional accuracy metrics with system-level indicators such as inference time per forecast update, end-to-end latency from data acquisition to prediction delivery, computational footprint, and operational robustness. Such reporting is essential for translating DL models from research benchmarks into reliable, real-world decision-support systems.

8.6 Benchmark datasets, standardization, and reproducibility

Although DL approaches for air quality predictions have been developed rapidly, meaningful comparison of different studies is challenging because of differences in data sources, data preprocessing options and evaluation protocols. Many investigations use different combinations of ground-based measurements, meteorological variables and products derived from satellite data, often processed with the help of study-specific assumptions. While this flexibility is beneficial to innovation, it also makes reproducibility more difficult and makes it hard to determine what was actually contributed by the design of a model. To remedy this gap, Table 11 summarizes data sources and methodological practices that seem to be found most frequently in the literature and may serve as a practical reference benchmark. The table describes in detail commonly used monitoring networks, reanalysis datasets, satellite Aerosol products, widely studied urban time series, along with commonly used preprocessing and validation strategies, such as missing data handling, feature normalization, temporal alignment, and time-aware evaluation. Greater consistency in the preparation and validation of data would enhance the comparability of data, and the cumulative value of DL-based air quality studies. Prioritizing publicly accessible data sets, reporting preprocessing steps and adopting chronological validation, standard forecast horizons and complementary perfor-

Table 11 Consolidated overview of benchmark datasets and methodological standards used in deep Learning–based air quality prediction studies

Aspect	Common Benchmarks/Practices	Standardization Recommendation
Air quality data	National/regional monitoring networks (CPCB, EPA AQS, CNEMC); frequently studied urban stations (Beijing, Delhi, Shanghai)	Report station coverage, temporal resolution, and missingness explicitly
Meteorological inputs	Reanalysis products (NCEP/NCAR, ERA5) and local weather observations	Ensure temporal alignment and consistent feature scaling
Satellite products	MODIS/MISR AOD and fused aerosol datasets	Document spatial resolution, quality filtering, and resampling methods
Preprocessing	Missing-value handling, normalization, lag construction	Use transparent imputation; report scaling method and lag windows
Validation	Chronological splits; rolling-origin evaluation	Avoid random shuffling; clearly define training and testing periods
Evaluation metrics	RMSE, MAE, R^2, normalized errors	Report multiple complementary metrics for fair comparison
Forecast tasks	Short-term (1–24 h) and multi-step prediction	Adopt common forecast horizons to enable benchmarking

mance measures may contribute to fairer comparison of models without limiting methodological innovation.

8.7 Context-dependent performance of DL models

Findings of the reviewed studies show that the performance of DL models in air quality forecasting is highly context-specific, depending on the type of pollutant, the spatial context, and the temporal resolution. No single architecture is consistently better than others in all situations, thus there is a need to match the choice of model with the nature of the data and underlying forecasting objectives. Recurrent models like LSTM and GRU are reported widely to produce good performance for particulate matter ($PM_{2.5}$ and PM_{10}) forecasting, especially at a resolution of hourly scale where there is a high degree of temporal persistence. In contrast, ozone (O_3) prediction, which is guided by nonlinear photochemical and meteorological interactions, frequently benefits from hybrid architectures that are designed to combine temporal learning with feature extraction. For pollutants with localised emission patterns such as NO_2 and SO_2, models that use spatial context and in particular in a well-monitored area in urban areas tend to show improved performance. Geographical and temporal contexts also determine how effective models are. Spatially aware models have obvious advantages in areas with dense coverage data (i.e., cities) while, in areas with limited monitoring coverage, more basic temporal models or transfer learning models are often competitive. Similarly, though complex DL architectures can be advantageous in cases of high resolution and multi-step forecasting, advantages become less evident at a coarse temporal scale in the future. These observations highlight the importance of context-aware interpretation of performance and the need for transparent justification of model choice, as opposed to generic claims of superiority of one architecture over the others.

9 Discussion

9.1 Limitations

While there is a broad coverage of DL studies for air quality prediction in this review, there are still a number of overarching challenges. One major problem is geographical bias and sampling bias: most studies use densely monitored urban areas, mainly China and India, with rural areas, coastal areas, mountainous areas and low-income areas being underrepresented, preventing the global applicability of studies. Publication bias also impacts the evidence base as research that provides evidence of good DL performance is more likely to be published. Reproducibility and comparability are also issues in progress. Many of these studies have little information about preprocessing steps, hyperparameter values, and training procedures, and this is combined with the dearth of open-source datasets. Differences in the characteristics of various datasets, evaluation practices, forecasting horizons, and performance measures make comparisons difficult. Despite the standardization attempts mentioned in this review, the absence of generally agreed benchmark datasets and common evaluation protocols is still limiting the accumulation of knowledge. Methodological weaknesses are a further limitation for the real-world applicability of the model. Cross validation strategies are often unclear, missing data are handled inconsistently and spatiotemporal heterogeneity is also not modeled sufficiently. Reported model superiority is often context specific - for example, it depends on the pollutant type, geography and temporal resolution - but in many cases, investigators make only generic claims about model performance, without critical and context-specific evaluation. Finally, the limited interpretability and a lack of ethical safeguards for the use of DL systems in public health and regulatory settings pose barriers to the adoption and use of DL for public health and its implementation in society, where the need for transparent, trustworthy and equitable models for informed decision-making is critical.

9.2 Future prospects

Addressing these limitations will go a long way to improve the maturity and practical relevance of DL-based air quality forecasting. Future progress will need further innovation in model design, such as hybrid DL frameworks, generative models, physics-informed networks and emerging paradigms (e.g. quantum neural systems) to support faster and more adaptive predictions under varied environmental conditions. It is equally important to have real-world integration. Combining DL models with IoT platforms and low-cost sensor networks with edge computing infrastructures can make fine-resolution real-time forecasting possible. However, such deployment must be done cautiously in terms of hardware limitations, data governance, scalability, and software maintenance. Strengthening the explainability of models is also crucial. Techniques such as SHAP, LIME, surrogates, attention visualization, and saliency mapping can provide transparency and help interpret pollution episodes or identify causality drivers and help communicate to regulatory stakeholders. Ethical and governance considerations, such as fairness, privacy, accountability and transparency, should be central to the system development process and underpinned by clear frameworks around data ownership, model oversight and lifecycle management. Improved quantification of uncertainty using probabilistic methods such as Bayesian neural networks,

ensemble modeling and Monte Carlo dropout will further facilitate the risk-aware decision-making process during high pollution events. Achieving meaningful standardization and global benchmarking will need harmonization of preprocessing pipelines, consistent evaluation metrics, train-test protocols, and multi-regional benchmark datasets. Together, these steps will enhance comparability, minimize geographical bias and enhance model fairness and reliability.

10 Conclusion

This review provides a comprehensive synthesis of recent advances in DL for air quality prediction, highlighting its growing significance for environmental monitoring, public health protection, and sustainable urban planning. By systematically evaluating standalone, hybrid, and spatiotemporal architectures across pollutants, forecasting horizons, and geographical contexts, we provide an integrated view of model performance, robustness, and applicability. Comparative analyses, including rank-based summaries, indicate that DL approaches generally outperform traditional statistical and deterministic models, particularly in capturing nonlinear interactions among multiple pollutants, meteorology, and atmospheric processes. Despite these advances, the literature exhibits substantial fragmentation due to heterogeneous datasets, inconsistent evaluation practices, regional concentration of studies, and limited reporting transparency. These factors restrict reproducibility, generalizability, and real-world applicability. Evidence remains biased toward data-rich urban areas, with minimal coverage of rural, coastal, mountainous, and low-income regions. Methodological inconsistencies, the scarcity of open-source resources, and insufficient handling of spatiotemporal heterogeneity further constrain operational deployment. Looking forward, DL has significant potential to transform air quality forecasting into a reliable decision-support tool. Realizing this potential will require coordinated efforts across scientific, technological, and governance dimensions. Key opportunities and directions include:

Research Agenda and Prioritized Recommendations:

- *Hybrid and Advanced Architectures*: Further development of hybrid DL frameworks, physics-informed networks, generative models, and emerging paradigms such as quantum neural systems is needed to improve predictive fidelity and adaptability under diverse environmental conditions.
- *Integration with IoT and Edge Platforms*: Combining DL models with IoT sensor networks, low-cost devices, and edge-computing infrastructure can enable fine-resolution, real-time forecasting. Deployment must address hardware limitations, sensor reliability, data governance, and software scalability to ensure practical utility.
- *Explainability*, *Uncertainty*, *and Ethical Safeguards*: Adoption of explainable AI (XAI) methods (e.g., SHAP, LIME, attention visualization) and robust uncertainty quantification will improve transparency, regulatory trust, and risk-aware decision-making. Concurrently, ethical safeguards—including fairness, privacy, accountability, and equitable deployment—must be embedded within governance frameworks.
- *Standardization and Benchmarking*: Harmonized preprocessing workflows, consistent evaluation metrics, uniform train-test protocols, and globally representative benchmark datasets are essential to enhance comparability, minimize geographical bias, and sup-

port cumulative evidence synthesis.

- *Context-Aware Model Selection*: Matching model architecture and design to specific pollutants, spatial settings, and temporal resolutions is critical. No single model excels universally, underscoring the importance of context-specific evaluation for operational decision-making.
- *Transparency and Open Science*: Promoting methodological rigor, clear reporting, and open access to data and code will accelerate reproducibility, facilitate cross-study comparisons, and foster equitable participation in global research.

By systematically pursuing these directions, the field can move toward resilient, actionable, and policy-relevant air quality forecasting systems. Such efforts will support climate resilience, informed environmental management, and the protection of population health, transforming DL from a research tool into an operational cornerstone of public-health and environmental policy.

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Declarations

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Authors and Affiliations

Salvator Lawrence[1] · Srimuruganandam Bhathmanabhan[2]

✉ Srimuruganandam Bhathmanabhan
bsrimuruganandam@vit.ac.in

[1] Department of Environmental and Water Resources Engineering, School of Civil Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India

[2] Centre for Clean Environment, Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India